Continuous coupling trajectories — run report

RUN

185740_cross-sweep_quick-test
Status: completed | Started: 2026-09-29T17:57:40.618474+00:00

DIAGNOSTIC SUMMARY

Target actuator(s): phi1; sine: 0.1164 Hz; measured PAX rate: 6.85 Hz; saved RP reference rate: 381 Hz; estimated PAX samples/cycle: 58.80.
Recorded cycles: 3; bias values: phi1 target 11, phi2 target 0. Waveform tracking: correct within descriptive threshold. Rising/falling branches (>5° descriptive threshold): strong separation present. Cycle repeatability (5° descriptive threshold): nonrepeatable above threshold. Bias-dependent plane changes (>5° descriptive threshold): obvious changes present. Low-DoP intervals: True. These thresholds are flags, not mechanism assignments.

PURPOSE

Continuously sine-drive phi1, phi2, or both while stepping and settling the orthogonal bias; record PAX plus measured OUT1/OUT2 references.

REQUIRED OPTICAL SETUP

PAX at final output; both paths open unless prompted. Wire OUT1 to IN1 and OUT2 to IN2 as voltage references. 'both' runs the complete diagnostic for both actuators automatically.

RECORDED BENCH CONTEXT (RECIPE)

In front of NPBS 1, C port
Red Pitaya (0-1 V command) -> Thorlabs MDT690 -> piezo

RUN COMMENT

cold run, haven't tested this tool update yet. this is less of a polarization test than just a dry run to make sure the update is preforming as intended.

OPERATOR SETUP NOTES

V_pi expectation only; verify channel and measured driver gain before calibration.

Continuous coupling trajectories — run report

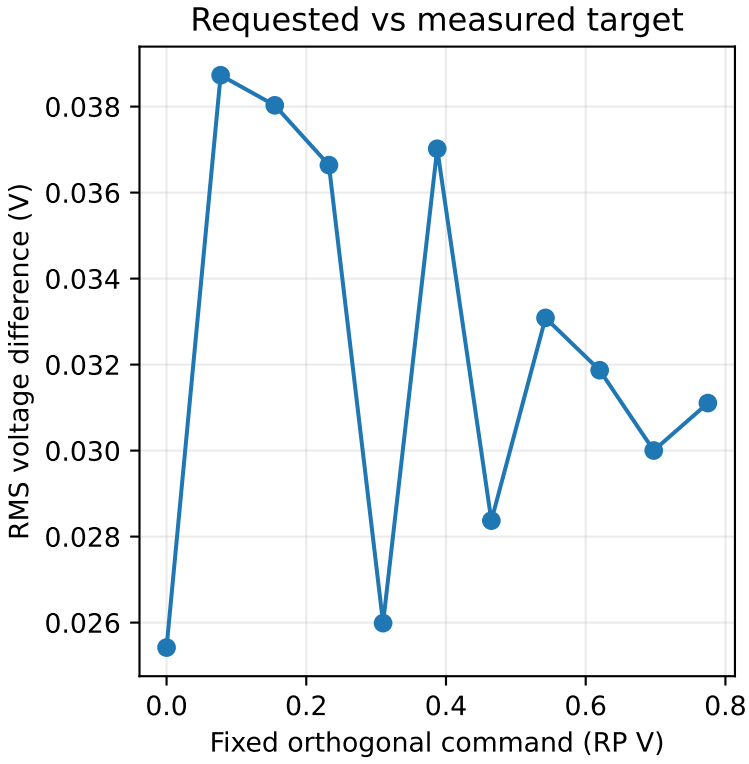
DATA

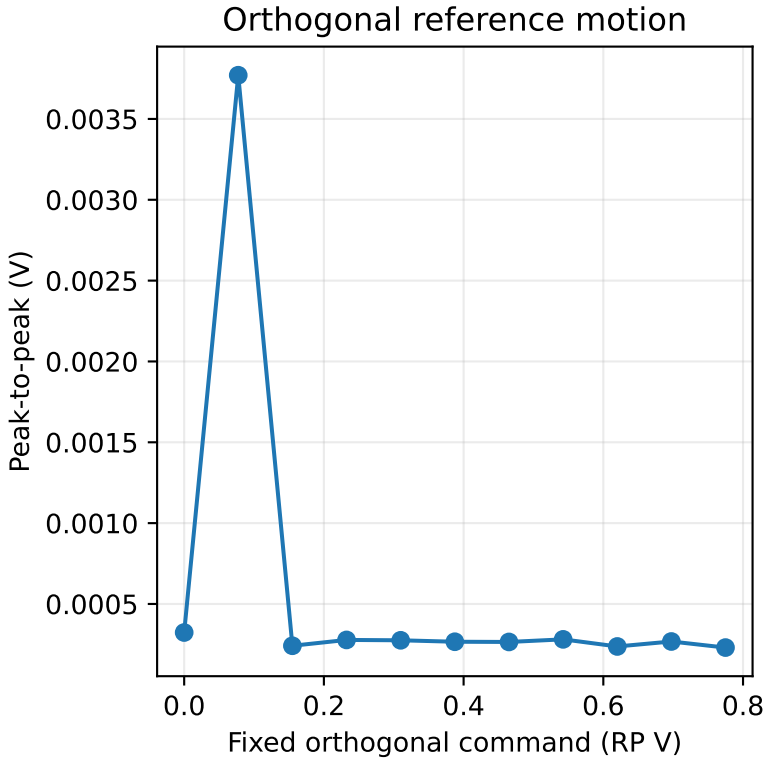
2,663 recorded samples; acquisition: 2,663
Source: data.csv; full acquisition settings: recipe.json; status: run.json

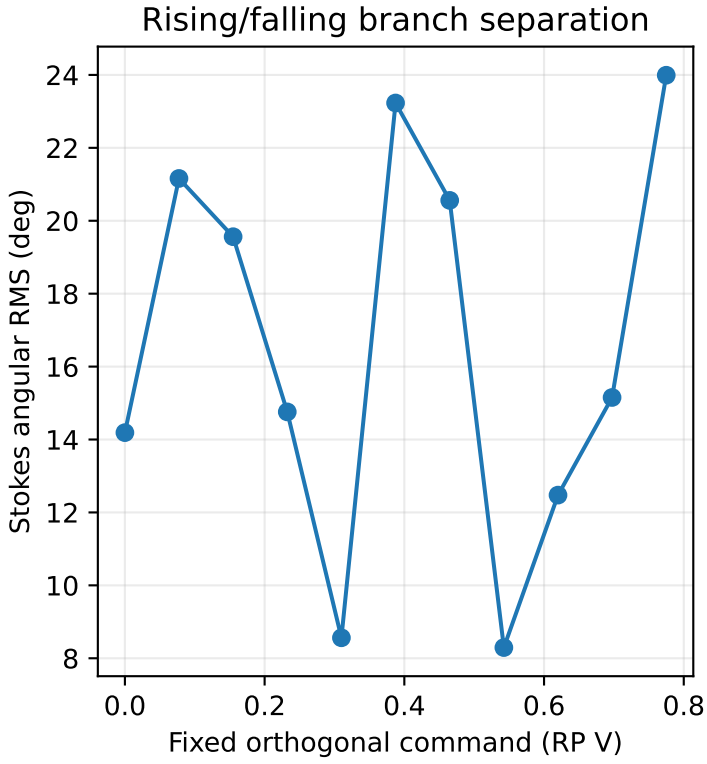
RAW DOP

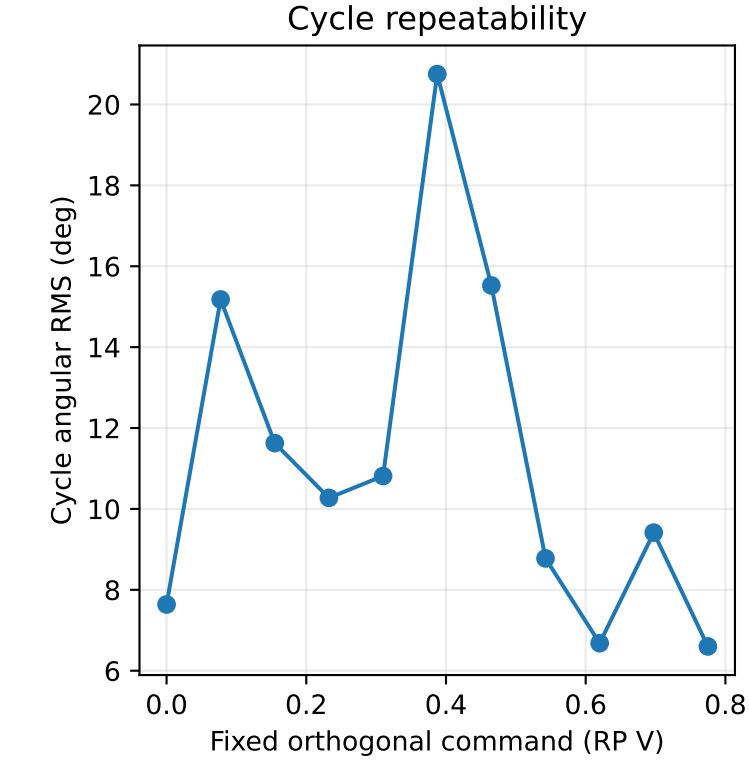
Mean 0.9218; range 0.3190 to 1.1744. 2 samples outside 0–1. Telemetry is not a new calibration.

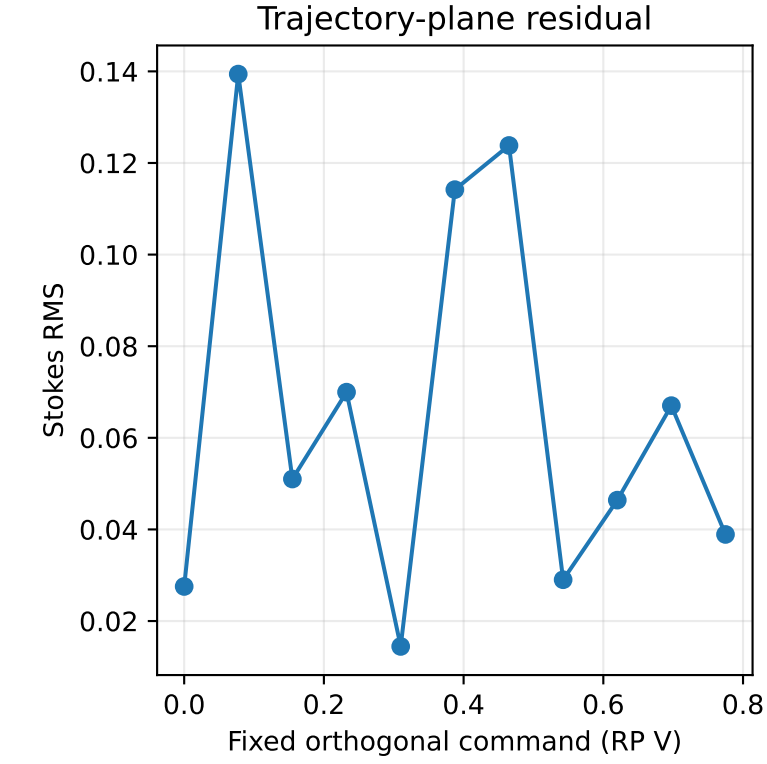
Continuous phi1 diagnostic — bias dependence

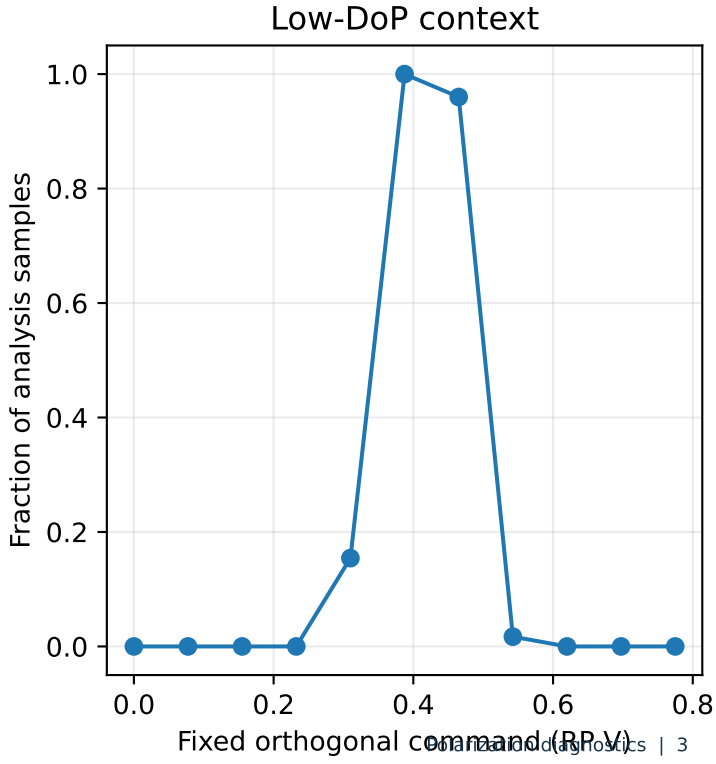




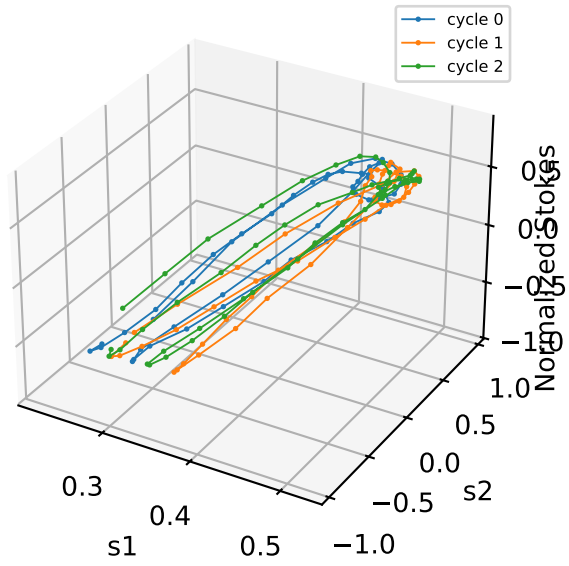


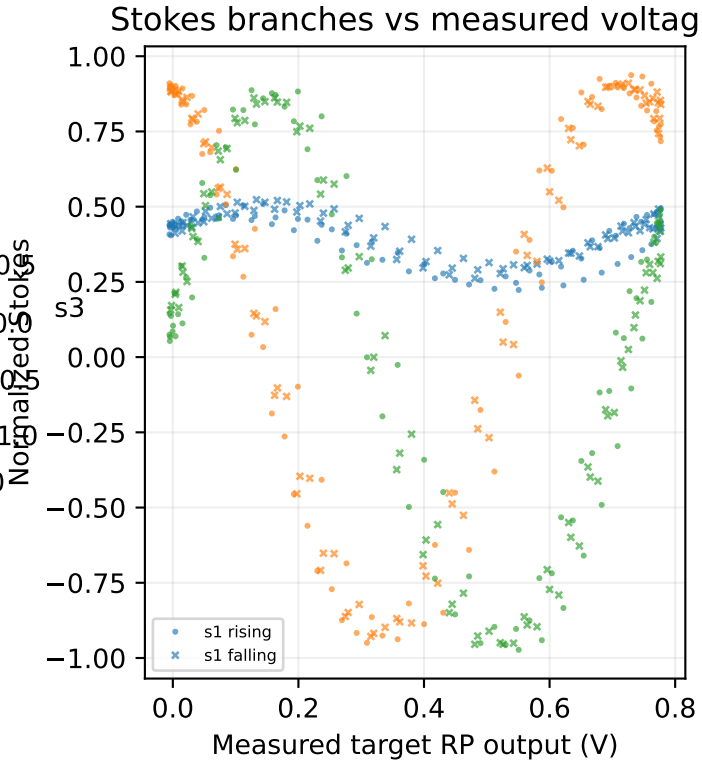


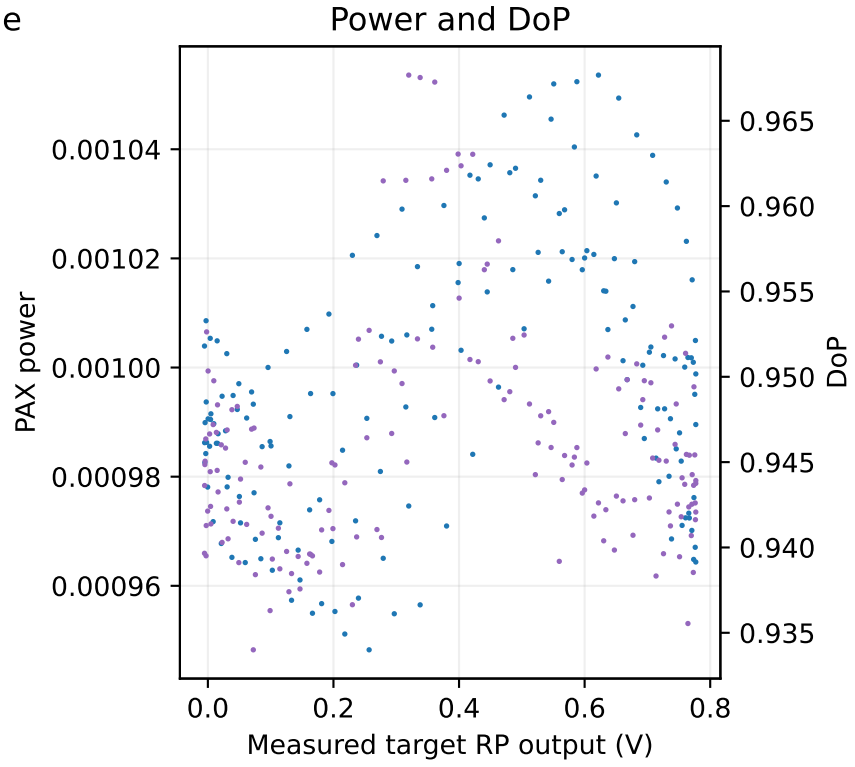


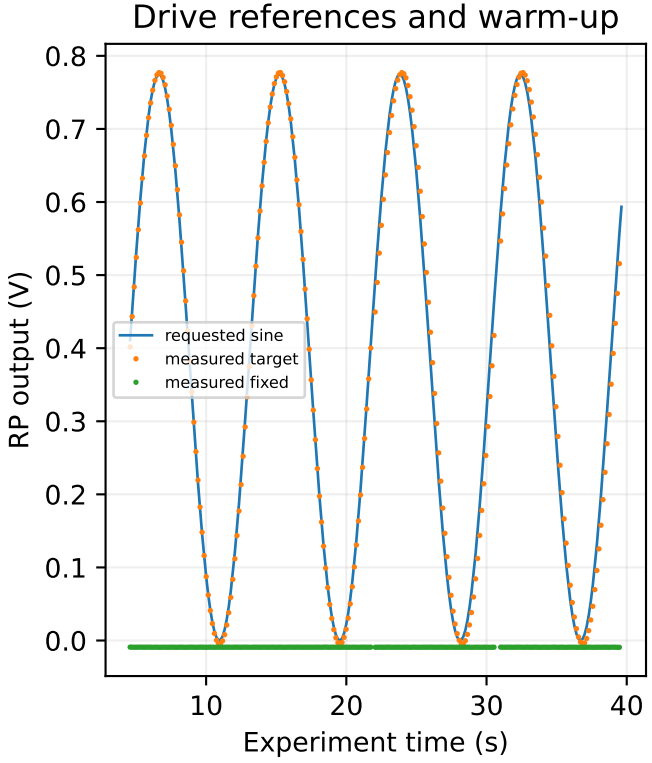


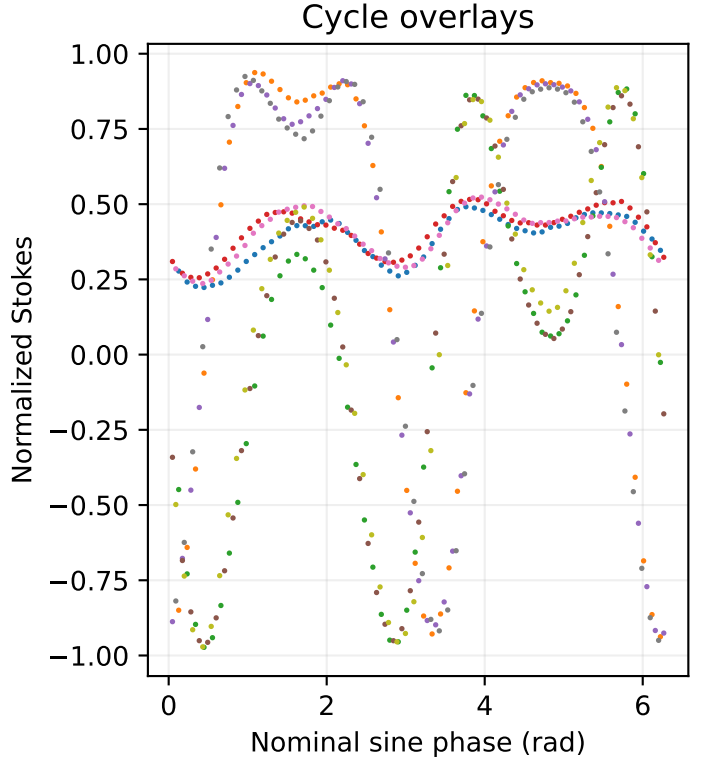
phi1 sine; fixed phi2 = 0 RP V (segment 0)





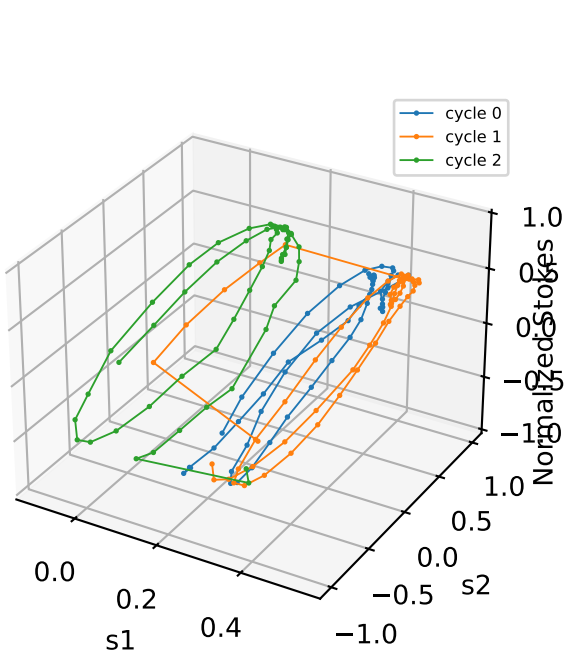


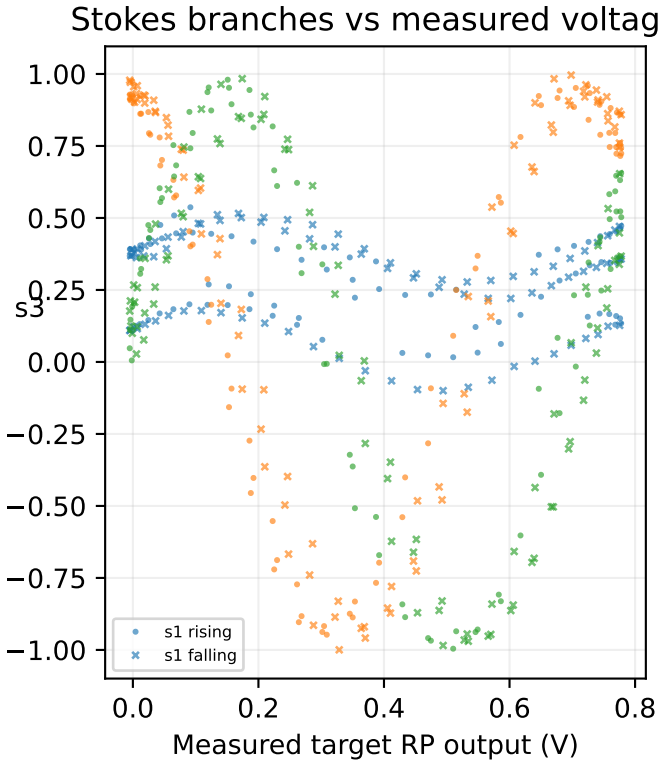


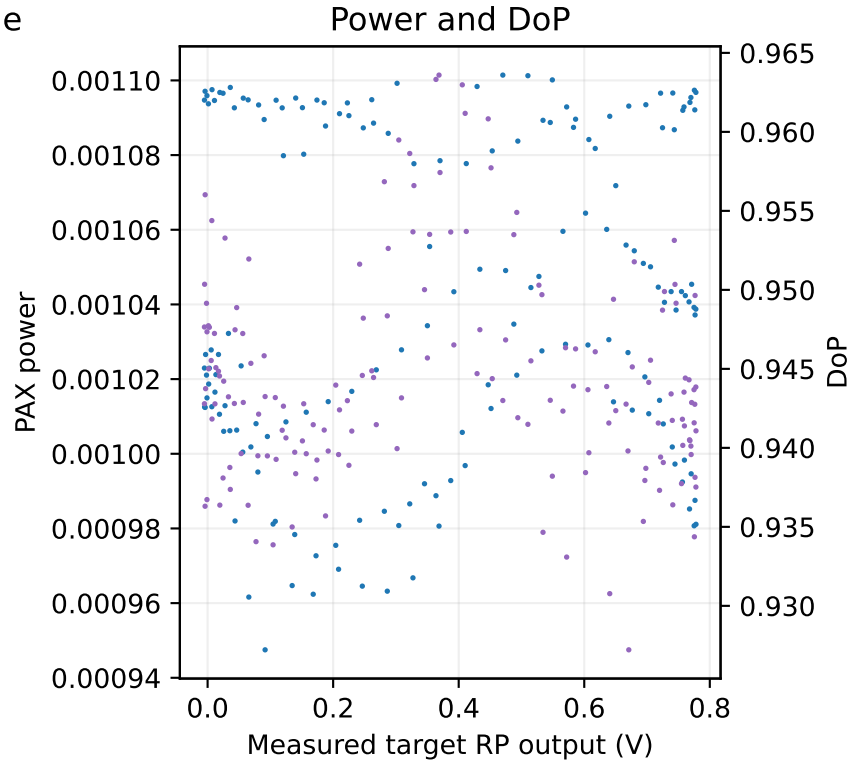


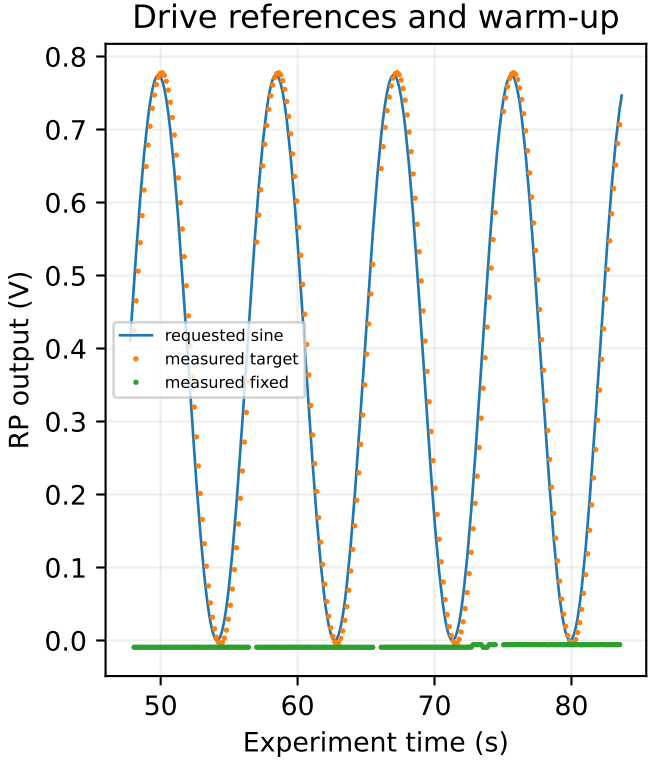
Ideal reference: phi1 should keep s1 fixed and total power independent of drive.
Plane normal: [0.992, -0.022, -0.125]
Plane residual RMS: 0.02755
Rising/falling angular RMS: 14.2°
Rising/falling power RMS: 3.18e-05 PAX units
Cycle angular RMS: 7.64°
Cycle power RMS: 2.12e-05 PAX units
Cycle DoP RMS: 0.00387
Tracking RMS: 0.02542 V
Reference coverage: 98.3%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.99, -0.02, -0.13], residual 0.026
Cycle 1 plane: [0.99, -0.02, -0.12], residual 0.0209
Cycle 2 plane: [0.99, -0.03, -0.12], residual 0.0269

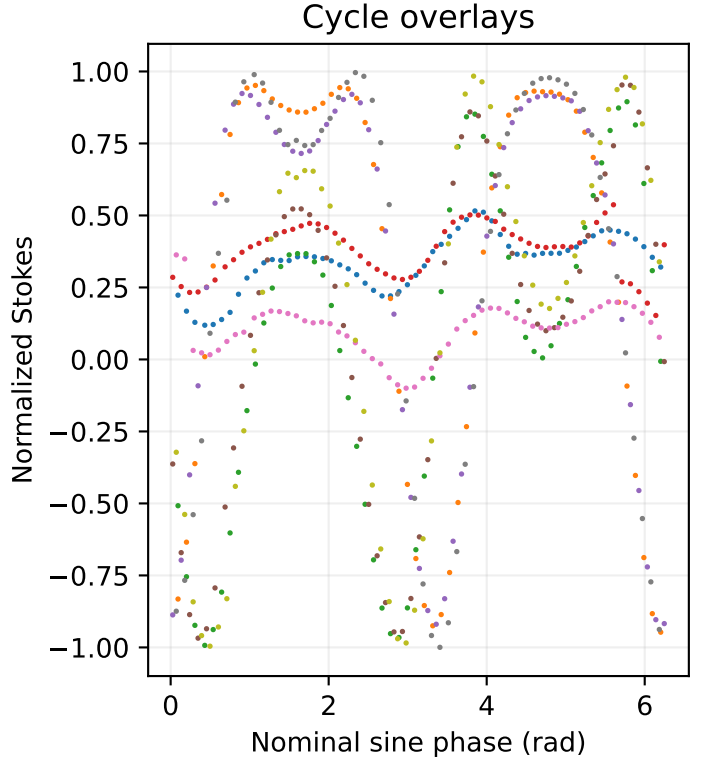
phi1 sine; fixed phi2 = 0.07739 RP V (segment 1)





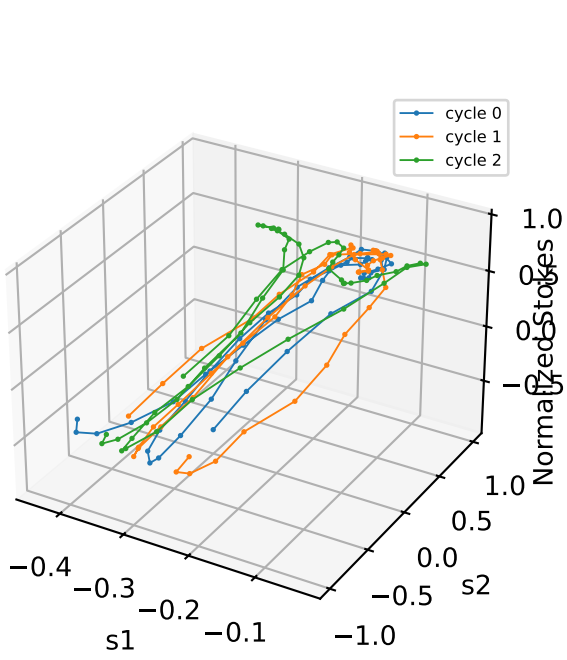


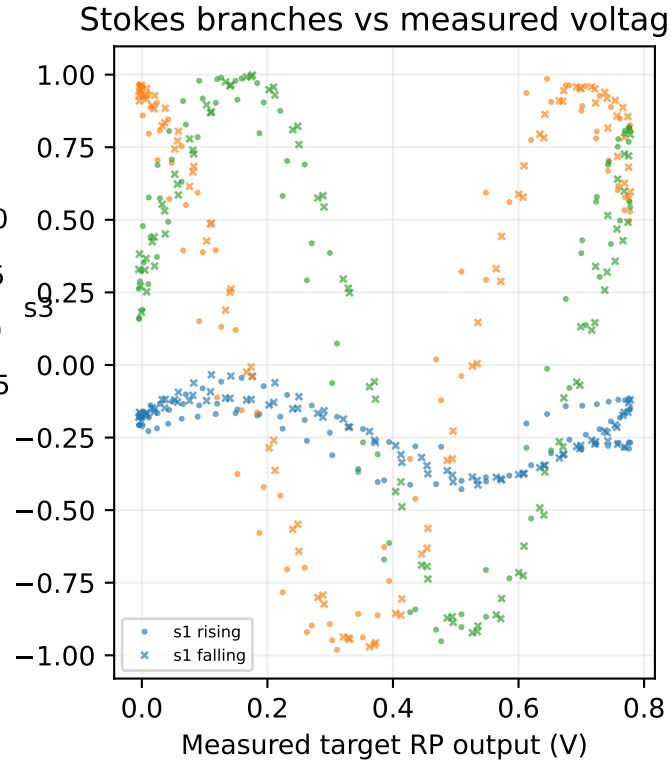


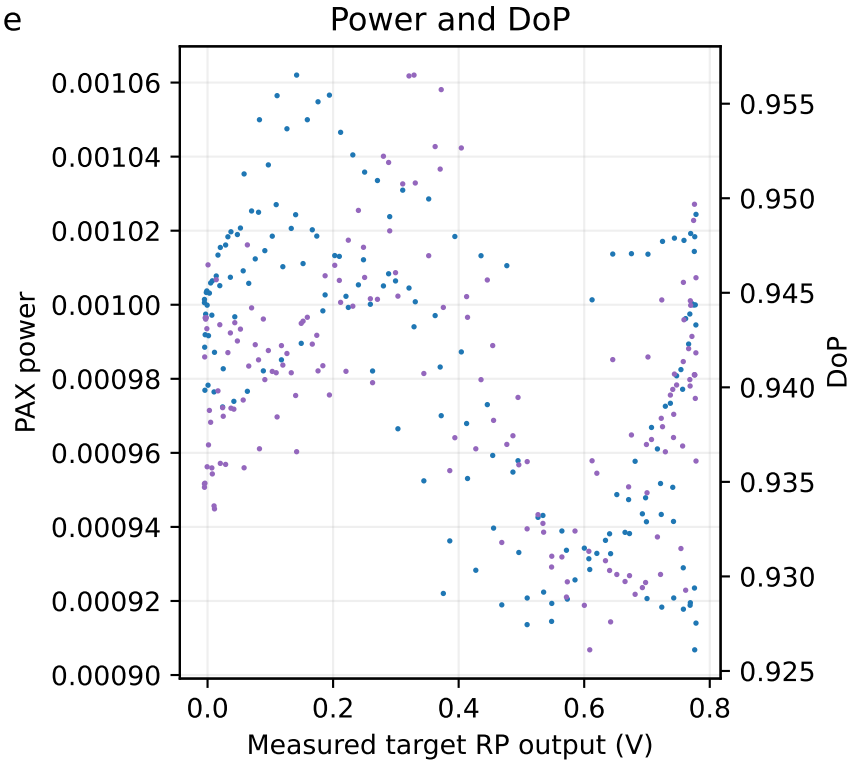


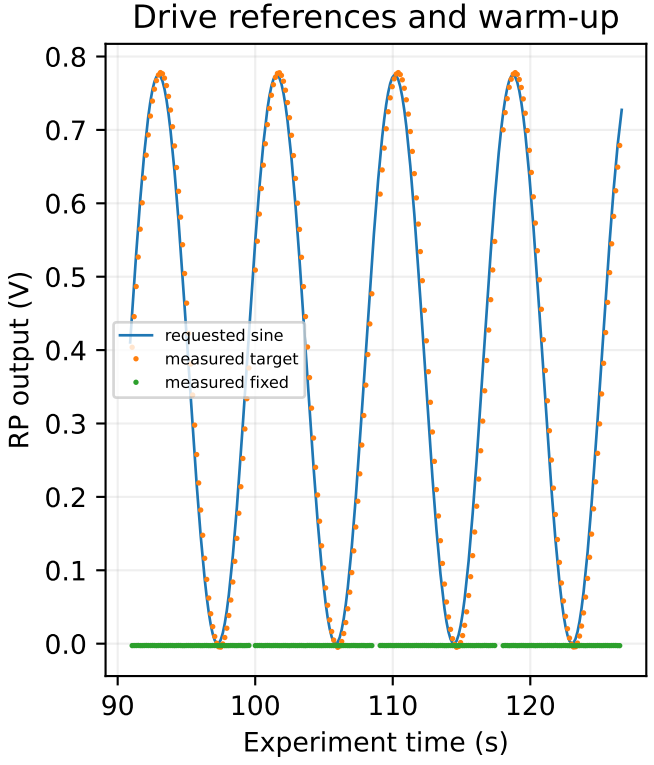
Ideal reference: phi1 should keep s1 fixed and total power independent of drive.
Plane normal: [0.995, -0.014, -0.103]
Plane residual RMS: 0.1394
Rising/falling angular RMS: 21.2°
Rising/falling power RMS: 4.29e-05 PAX units
Cycle angular RMS: 15.2°
Cycle power RMS: 5.55e-05 PAX units
Cycle DoP RMS: 0.00547
Tracking RMS: 0.03873 V
Reference coverage: 94.9%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.99, 0.01, -0.16], residual 0.0349
Cycle 1 plane: [1.00, -0.05, -0.09], residual 0.0653
Cycle 2 plane: [1.00, -0.00, -0.09], residual 0.0701

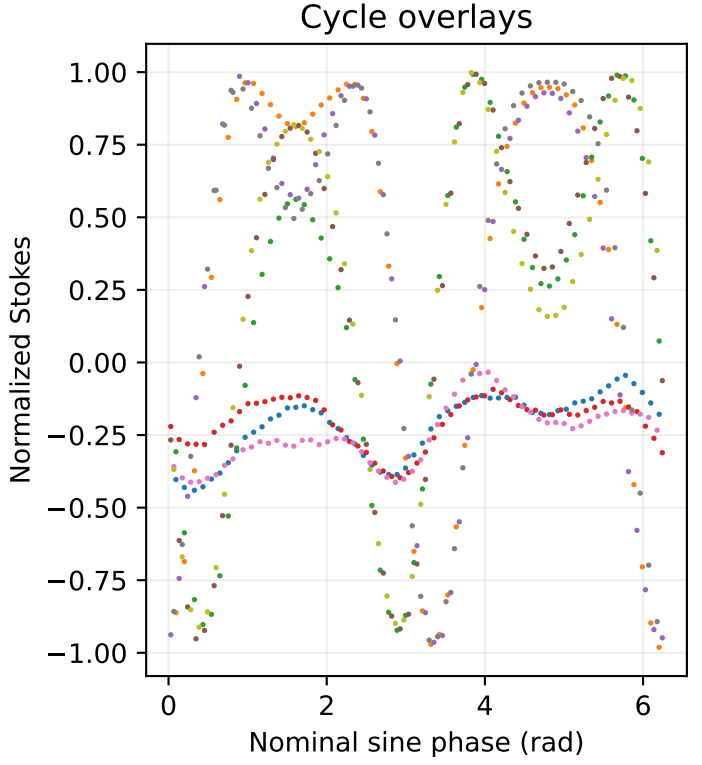
phi1 sine; fixed phi2 = 0.1549 RP V (segment 2)





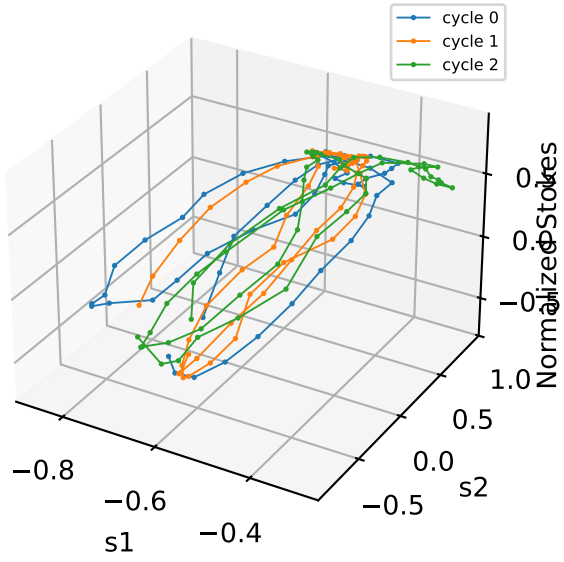


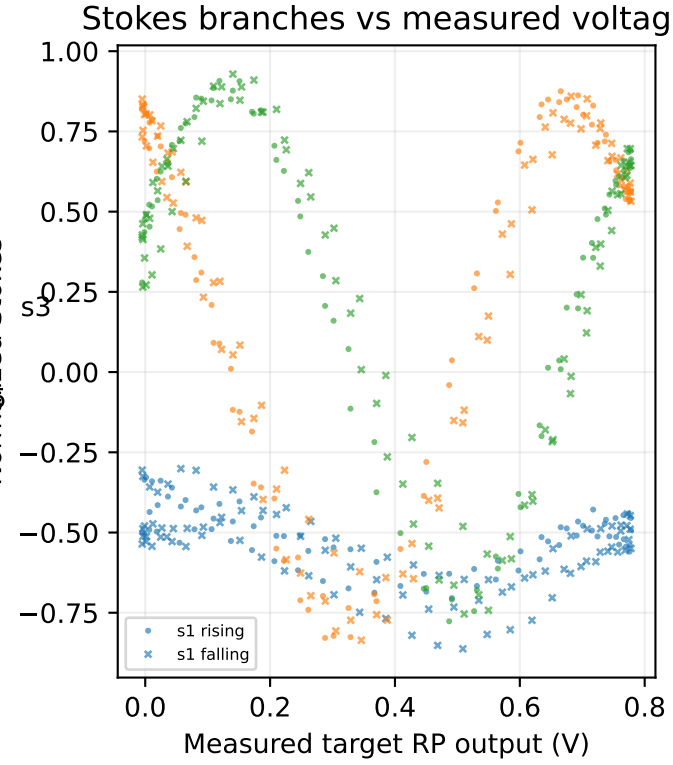


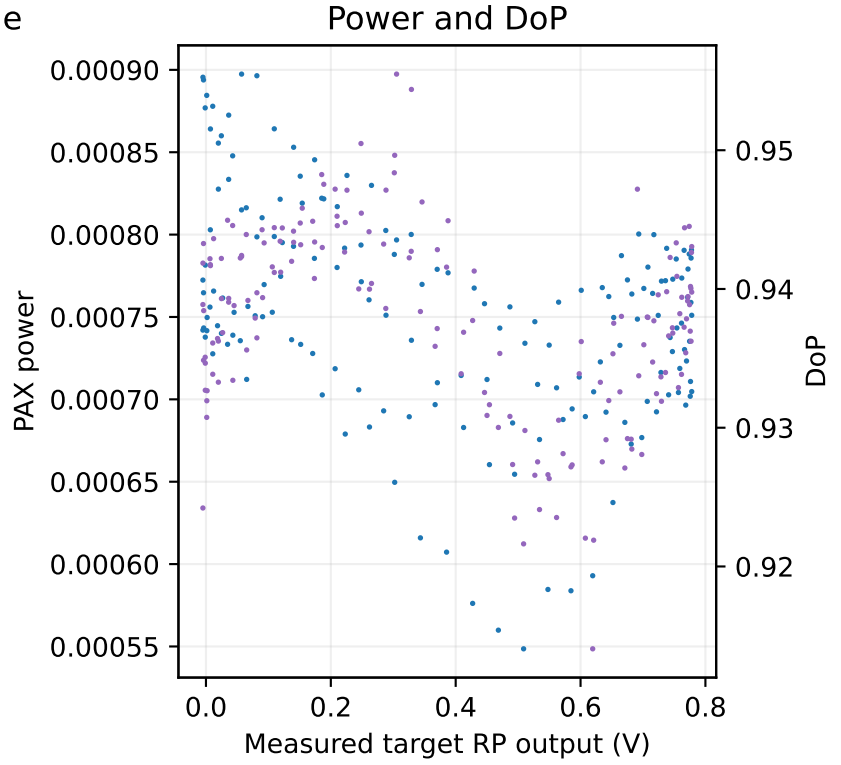


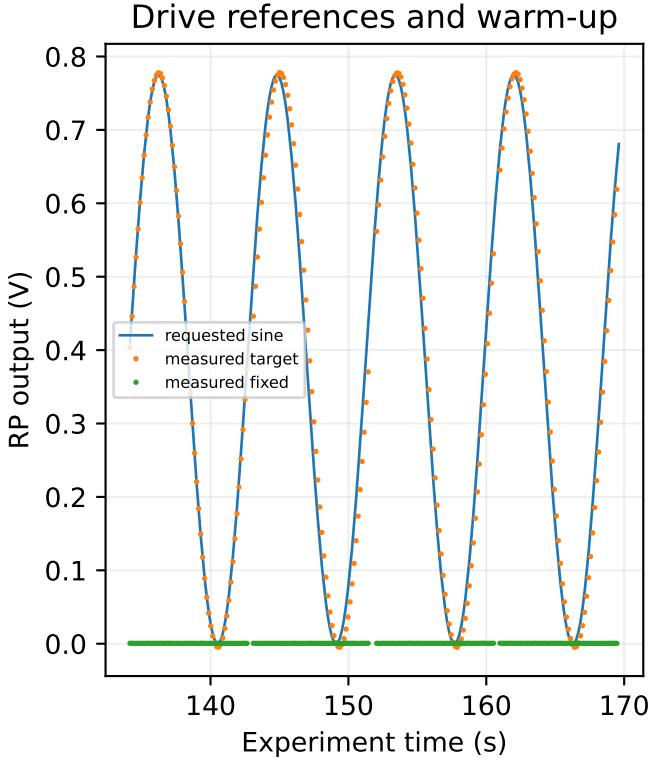
Ideal reference: phi1 should keep s1 fixed and total power independent of drive.
Plane normal: [0.990, -0.003, -0.141]
Plane residual RMS: 0.051
Rising/falling angular RMS: 19.6°
Rising/falling power RMS: 3.83e-05 PAX units
Cycle angular RMS: 11.6°
Cycle power RMS: 4.03e-05 PAX units
Cycle DoP RMS: 0.00421
Tracking RMS: 0.03803 V
Reference coverage: 95.4%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.98, 0.00, -0.17], residual 0.0305
Cycle 1 plane: [0.99, -0.02, -0.11], residual 0.0375
Cycle 2 plane: [0.99, 0.01, -0.14], residual 0.0599

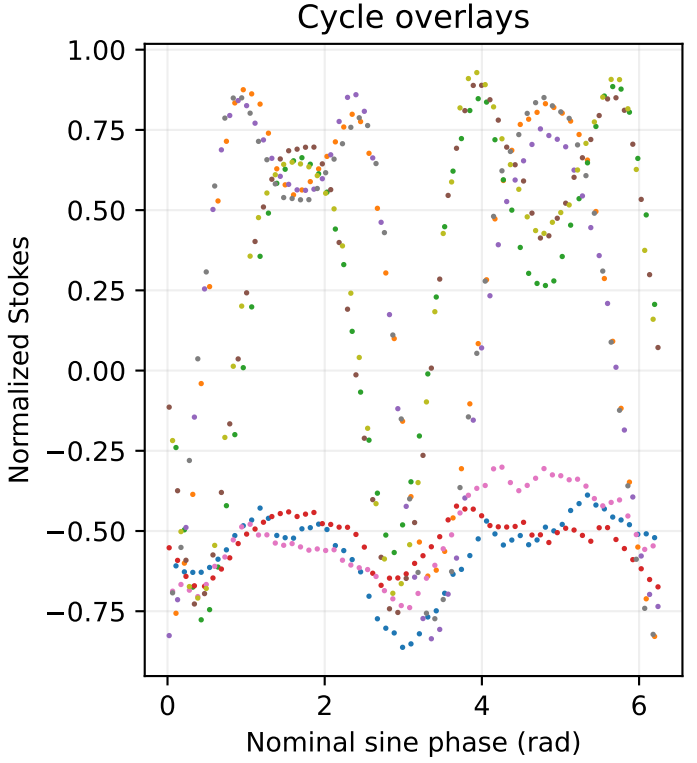
phi1 sine; fixed phi2 = 0.2324 RP V (segment 3)





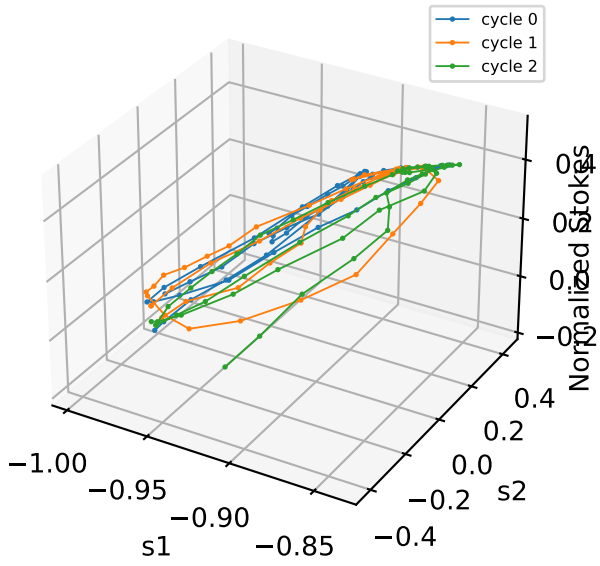


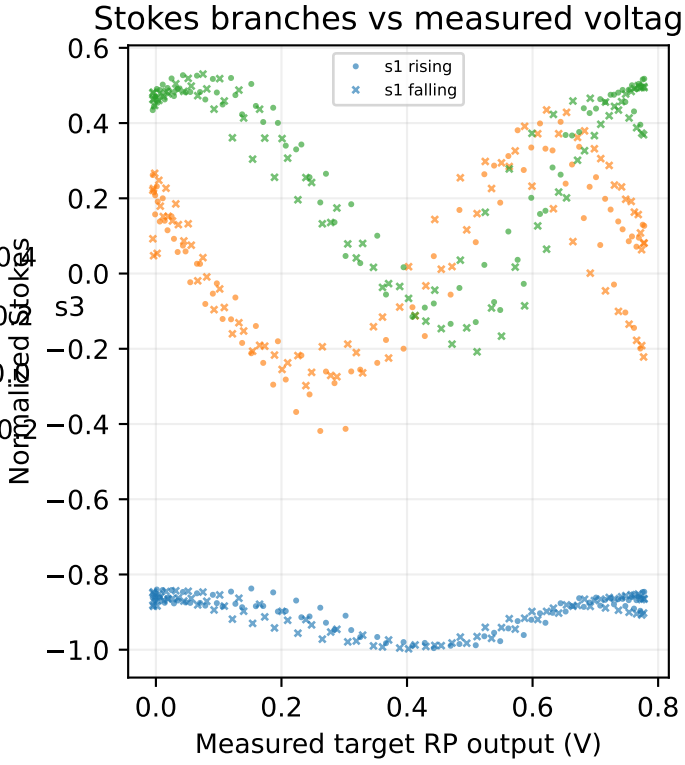


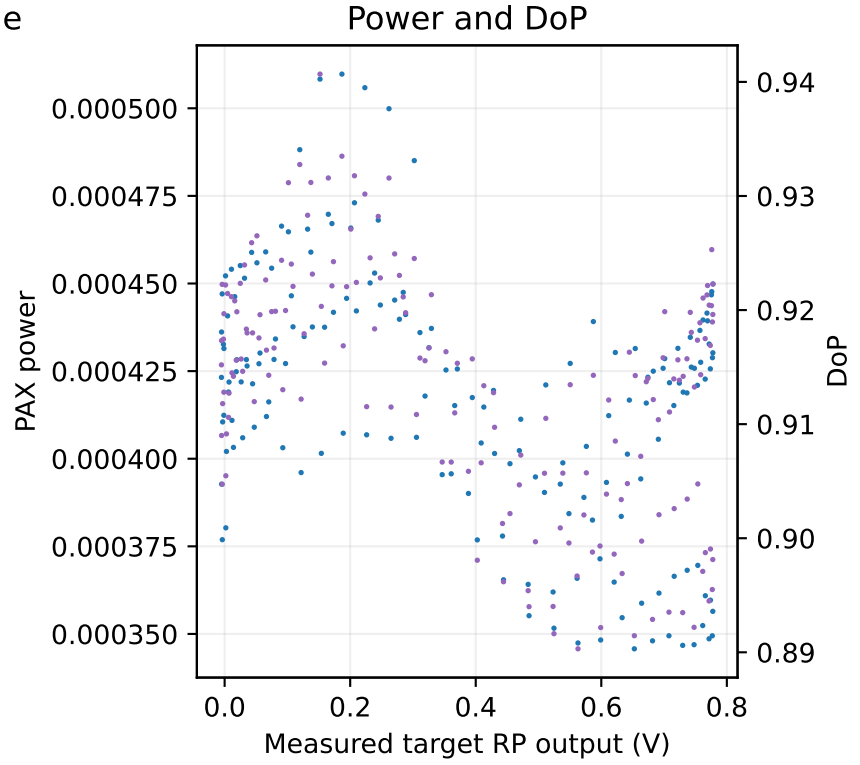


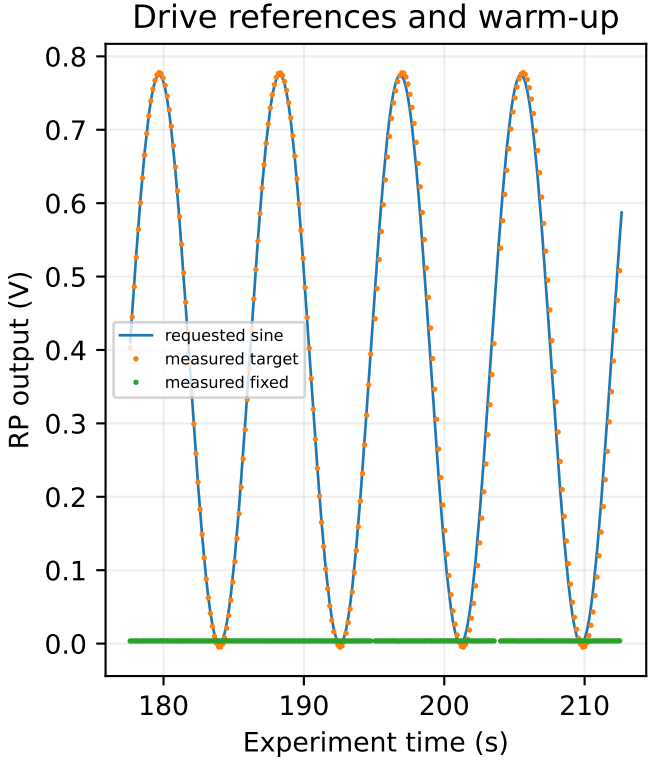
Ideal reference: phi1 should keep s1 fixed and total power independent of drive.
Plane normal: [0.987, -0.053, -0.151]
Plane residual RMS: 0.06997
Rising/falling angular RMS: 14.8°
Rising/falling power RMS: 8.85e-05 PAX units
Cycle angular RMS: 10.3°
Cycle power RMS: 8.39e-05 PAX units
Cycle DoP RMS: 0.00431
Tracking RMS: 0.03664 V
Reference coverage: 96.0%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.99, -0.08, -0.15], residual 0.0754
Cycle 1 plane: [0.99, -0.02, -0.11], residual 0.0396
Cycle 2 plane: [0.98, -0.06, -0.19], residual 0.0694

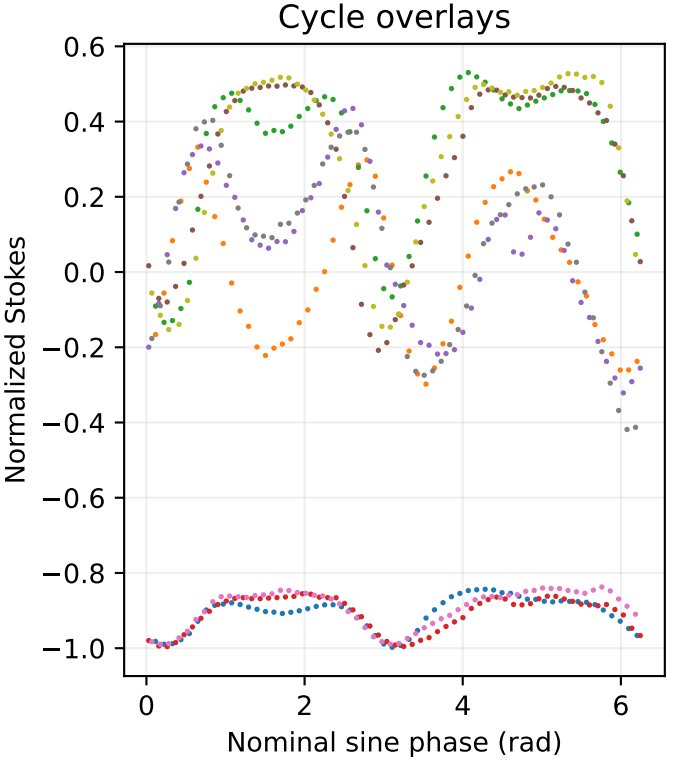
phi1 sine; fixed phi2 = 0.3099 RP V (segment 4)





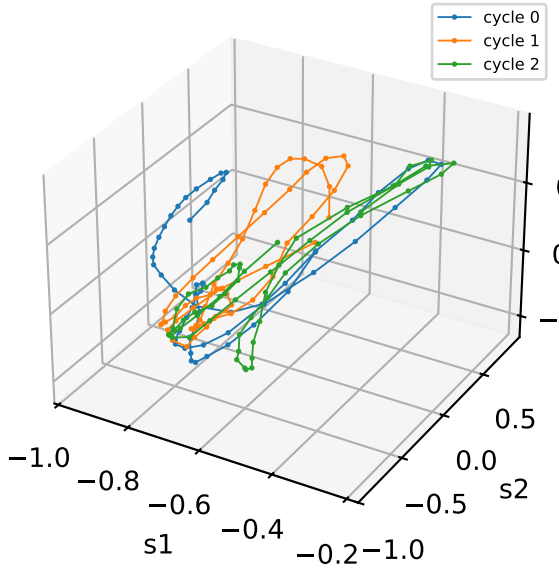


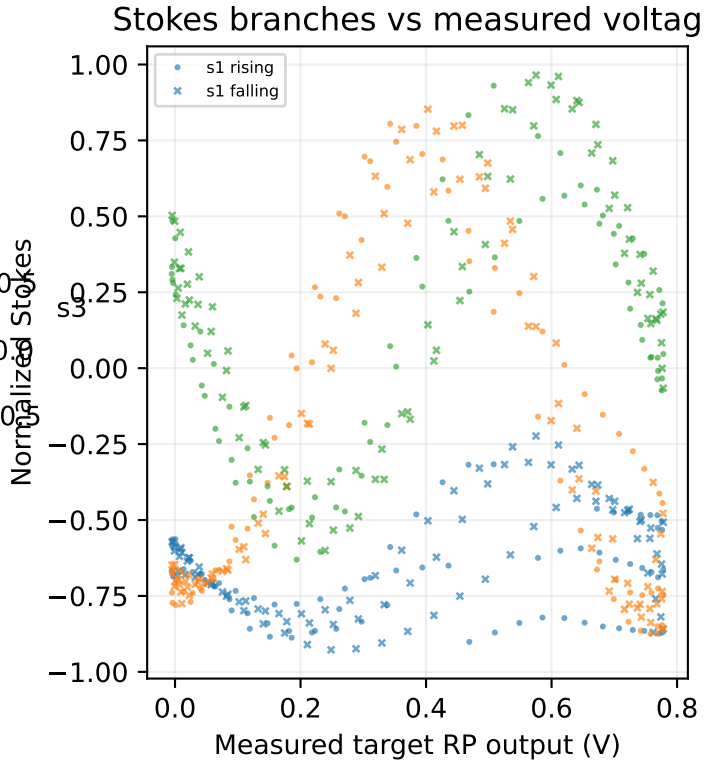


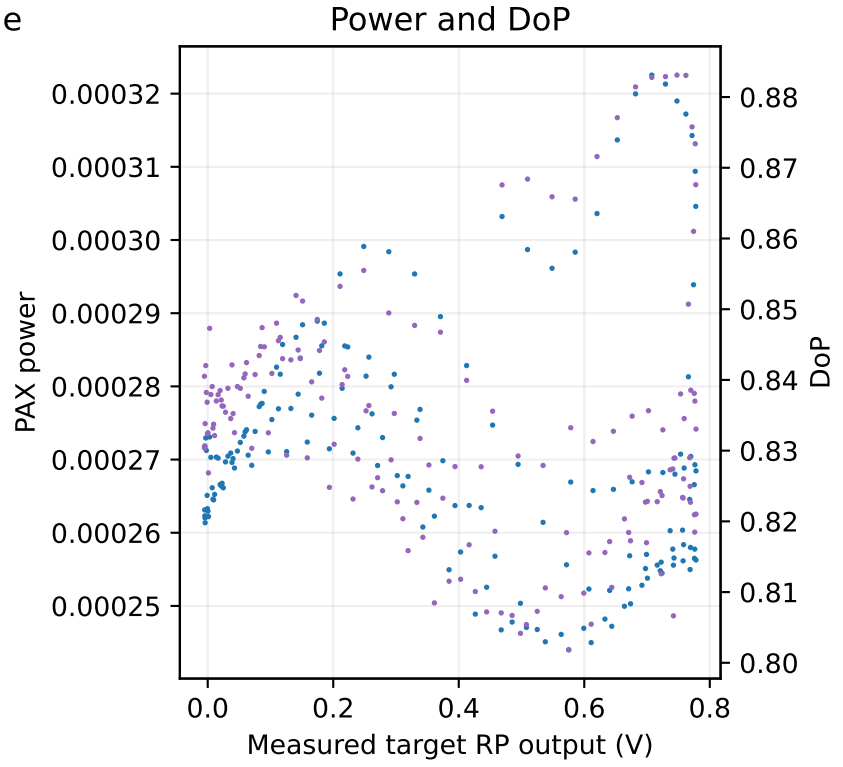


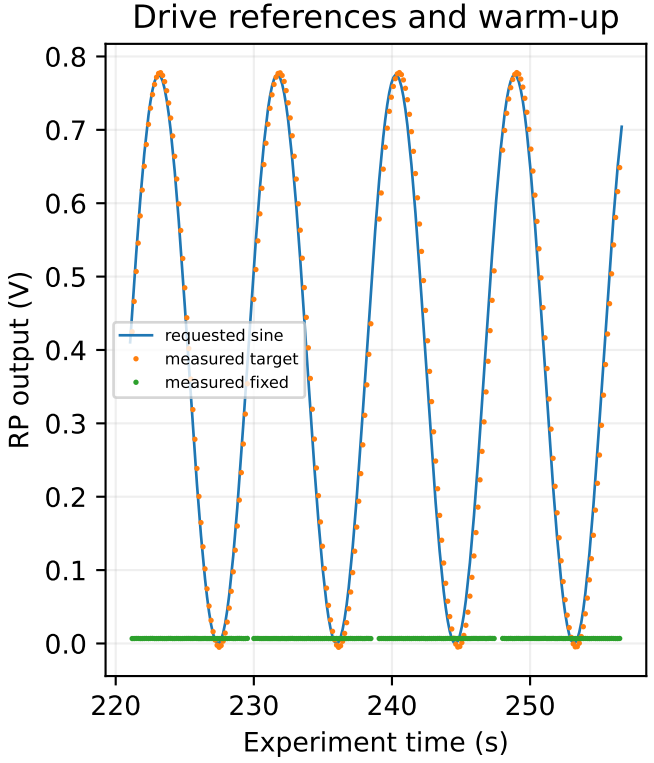
Ideal reference: phi1 should keep s1 fixed and total power independent of drive.
Plane normal: [0.980, -0.047, -0.192]
Plane residual RMS: 0.01446
Rising/falling angular RMS: 8.56°
Rising/falling power RMS: 3.64e-05 PAX units
Cycle angular RMS: 10.8°
Cycle power RMS: 4.92e-05 PAX units
Cycle DoP RMS: 0.0127
Tracking RMS: 0.02599 V
Reference coverage: 98.3%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.98, -0.04, -0.21], residual 0.00911
Cycle 1 plane: [0.98, -0.07, -0.18], residual 0.0136
Cycle 2 plane: [0.98, -0.02, -0.20], residual 0.0135

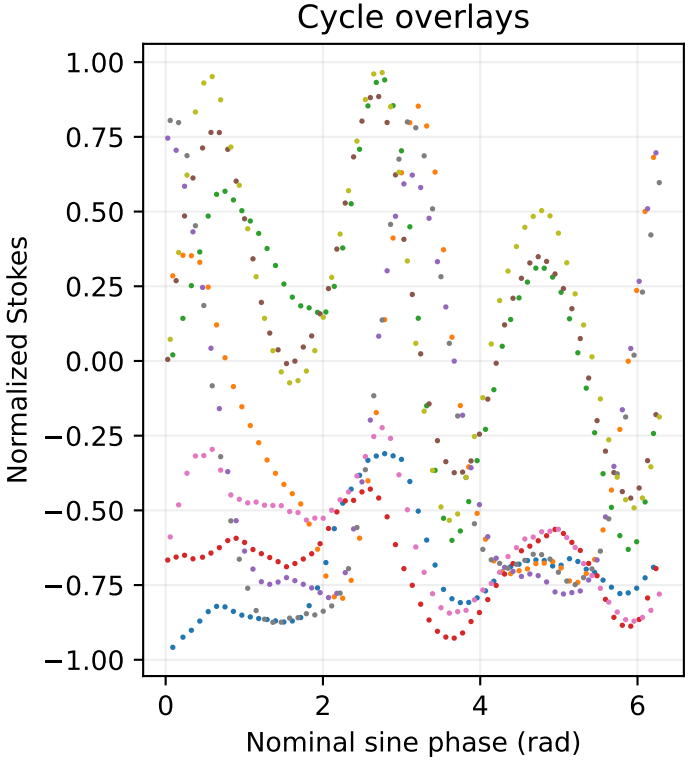
phi1 sine; fixed phi2 = 0.3875 RP V (segment 5)





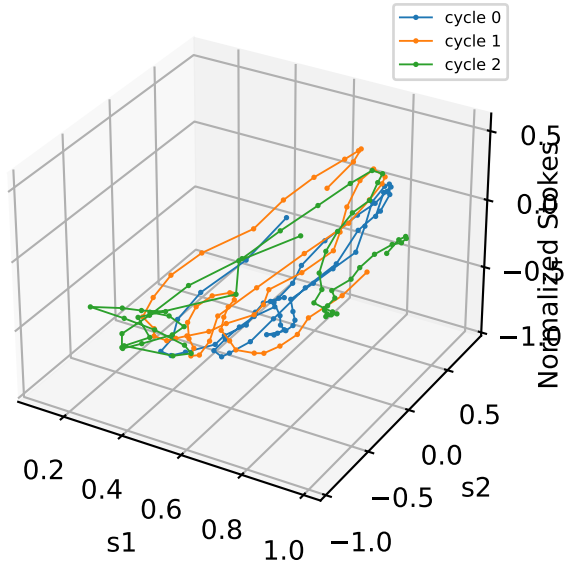


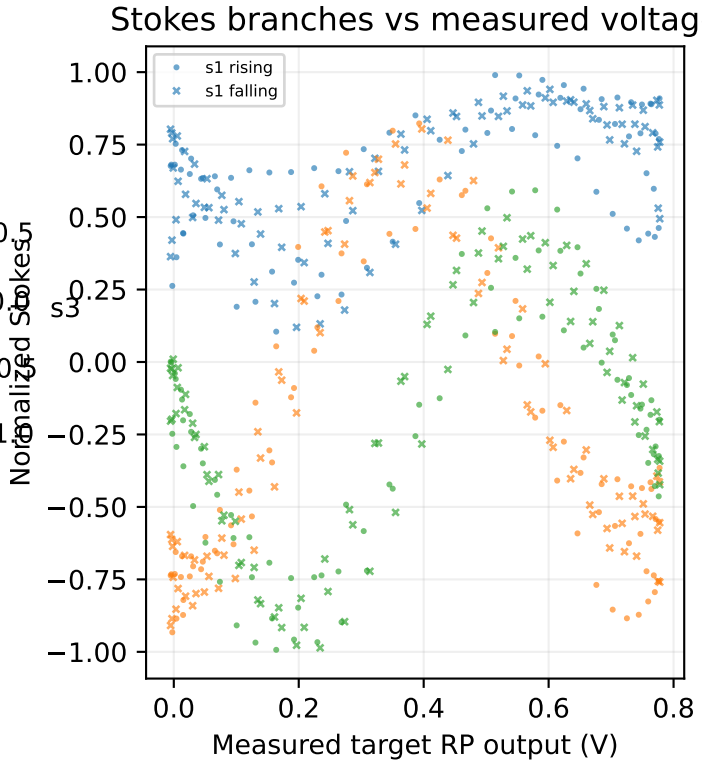


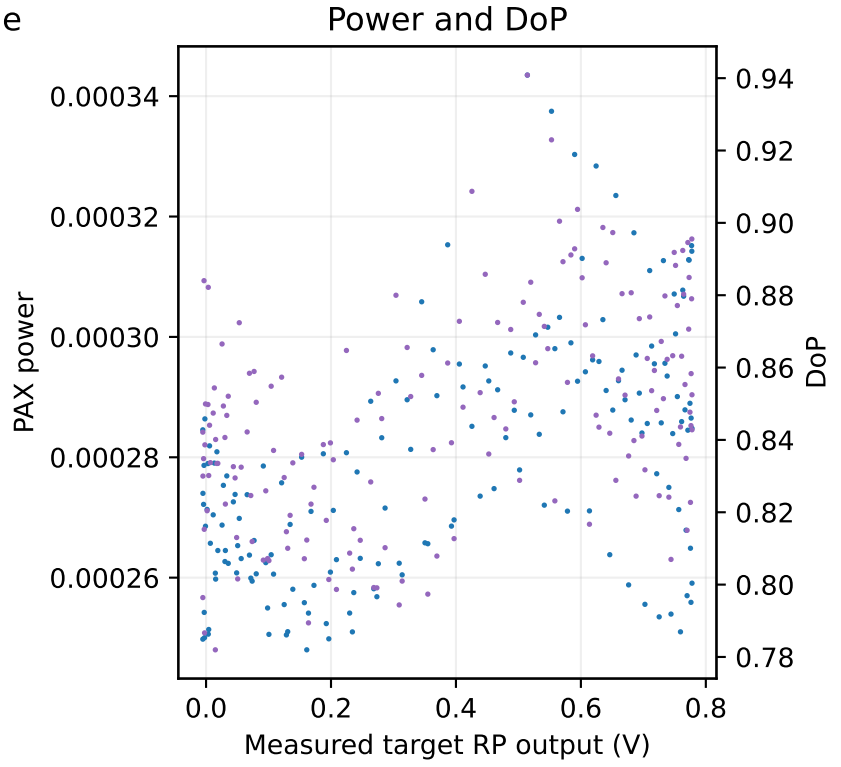


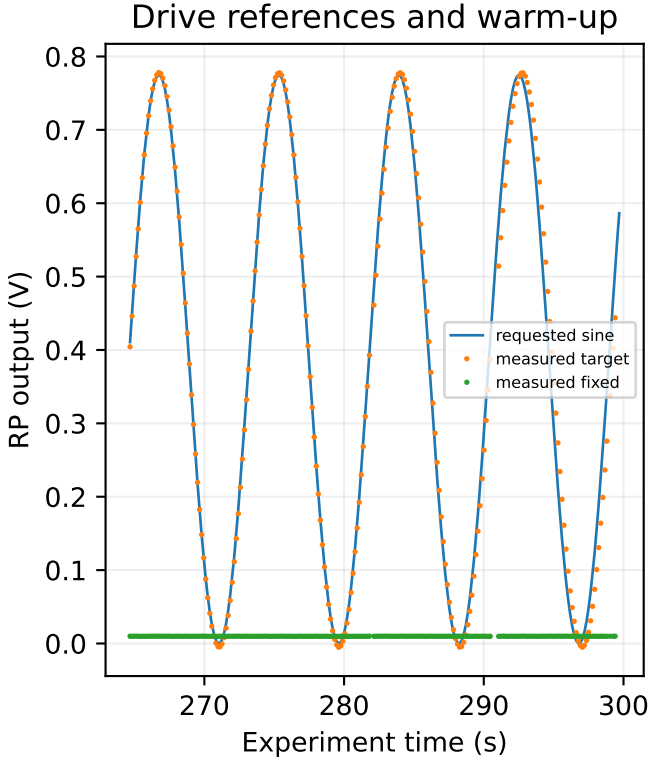
Ideal reference: phi1 should keep s1 fixed and total power independent of drive.
Plane normal: [0.954, 0.018, -0.298]
Plane residual RMS: 0.1142
Rising/falling angular RMS: 23.2°
Rising/falling power RMS: 2.36e-05 PAX units
Cycle angular RMS: 20.8°
Cycle power RMS: 2.91e-05 PAX units
Cycle DoP RMS: 0.0296
Tracking RMS: 0.03702 V
Reference coverage: 95.5%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.98, -0.03, -0.20], residual 0.148
Cycle 1 plane: [0.96, 0.07, -0.28], residual 0.053
Cycle 2 plane: [0.93, -0.00, -0.36], residual 0.0724

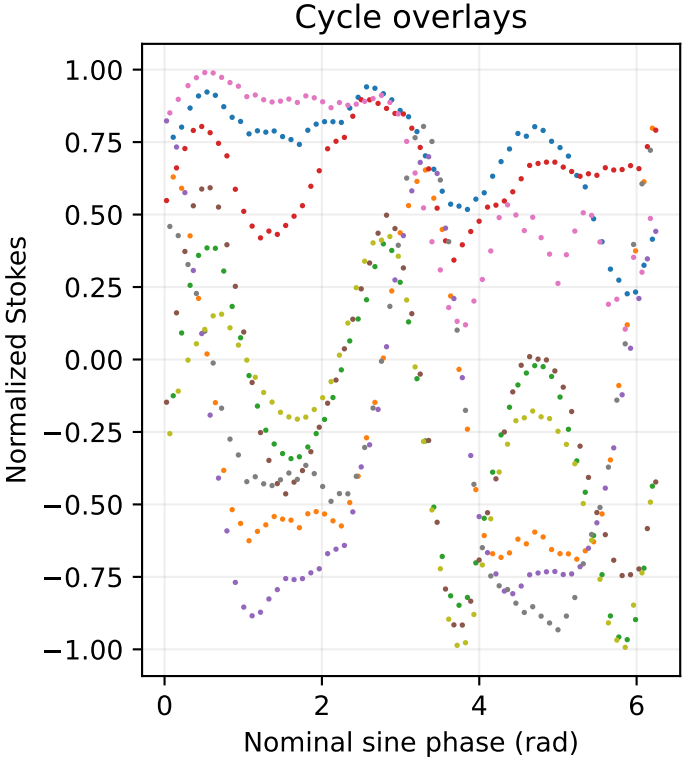
phi1 sine; fixed phi2 = 0.465 RP V (segment 6)





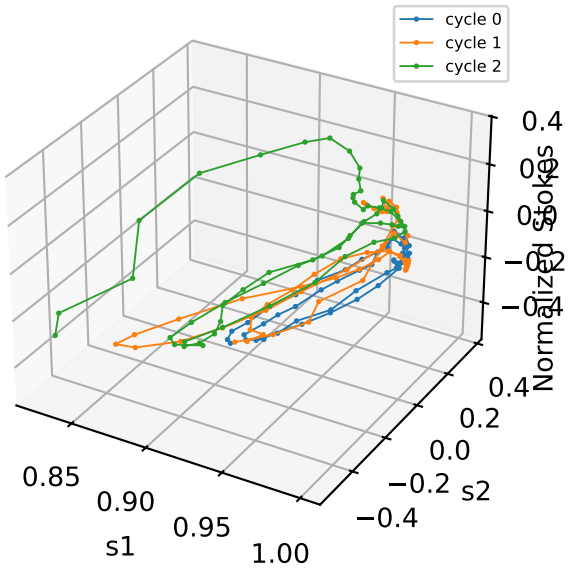


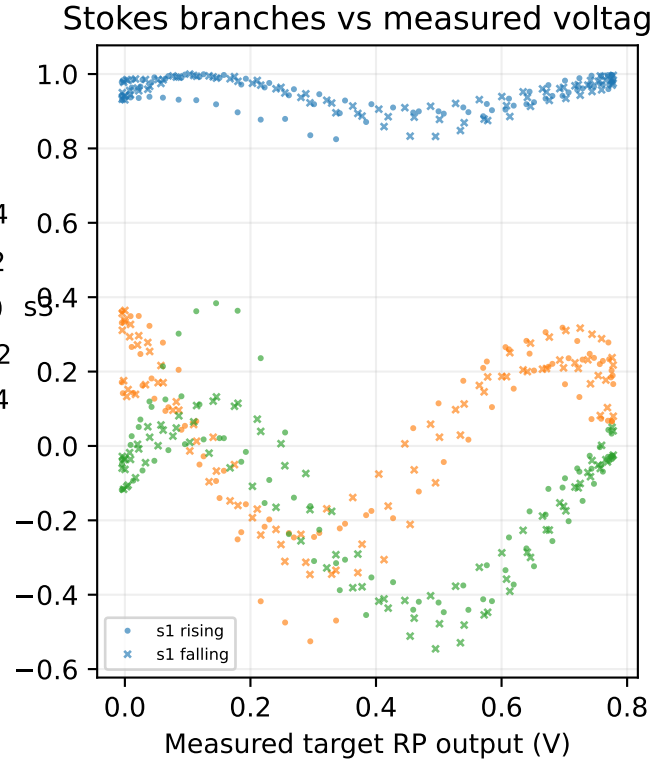


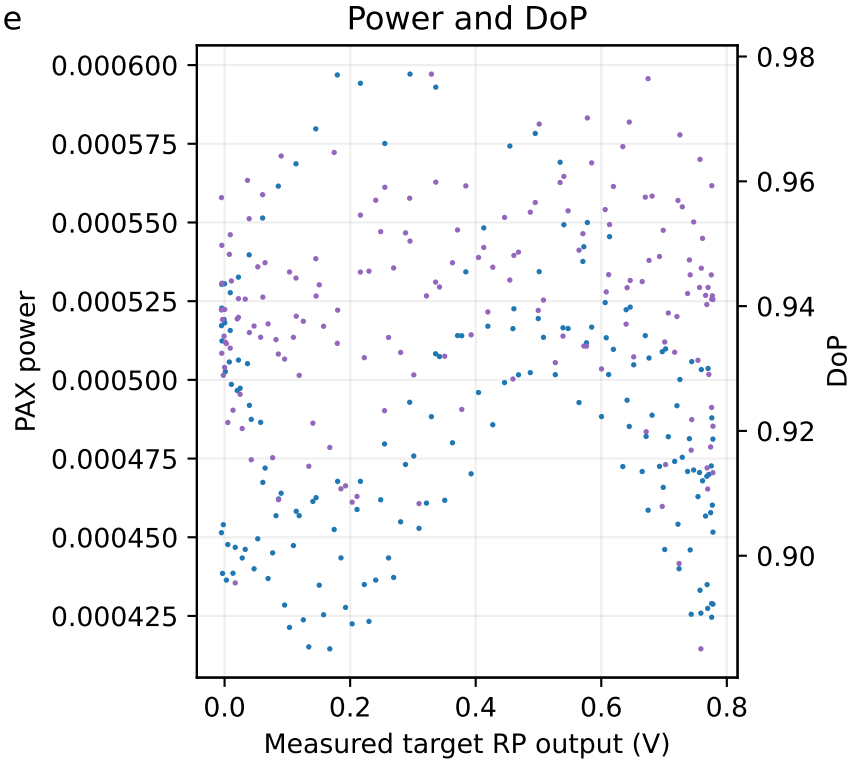


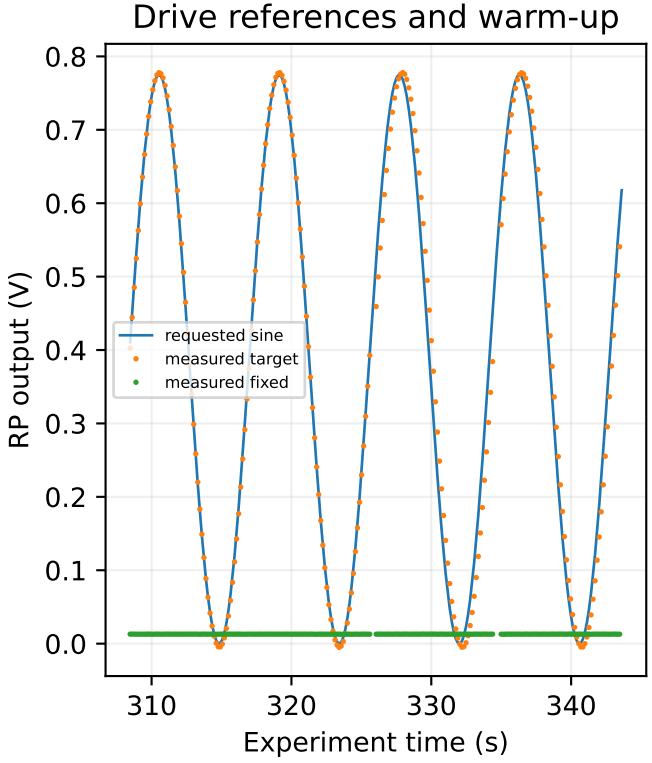
Ideal reference: phi1 should keep s1 fixed and total power independent of drive.
Plane normal: [0.906, -0.020, -0.423]
Plane residual RMS: 0.1238
Rising/falling angular RMS: 20.6°
Rising/falling power RMS: 2.2e-05 PAX units
Cycle angular RMS: 15.5°
Cycle power RMS: 2.12e-05 PAX units
Cycle DoP RMS: 0.0329
Tracking RMS: 0.02837 V
Reference coverage: 97.7%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.91, 0.06, -0.41], residual 0.0639
Cycle 1 plane: [0.97, -0.06, -0.23], residual 0.0927
Cycle 2 plane: [0.82, -0.04, -0.57], residual 0.123

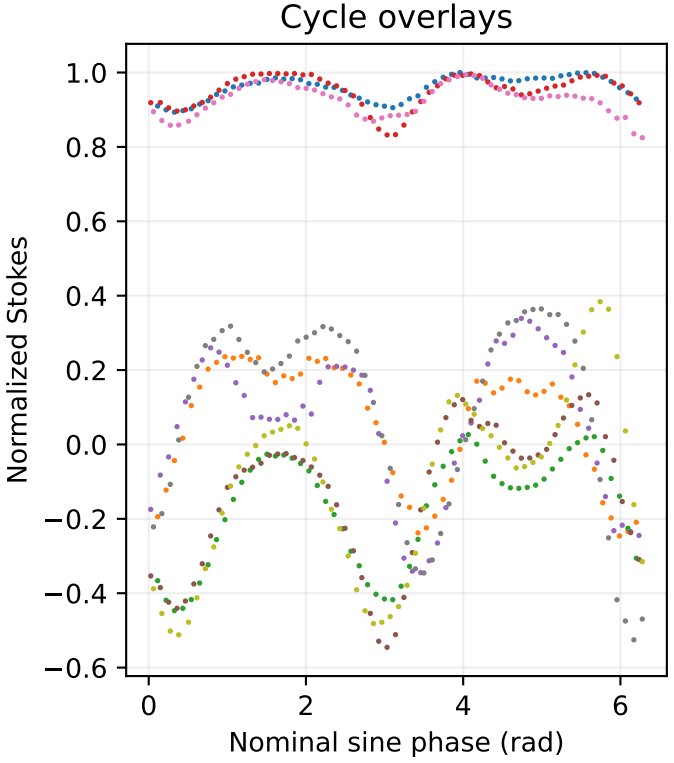
phi1 sine; fixed phi2 = 0.5425 RP V (segment 7)





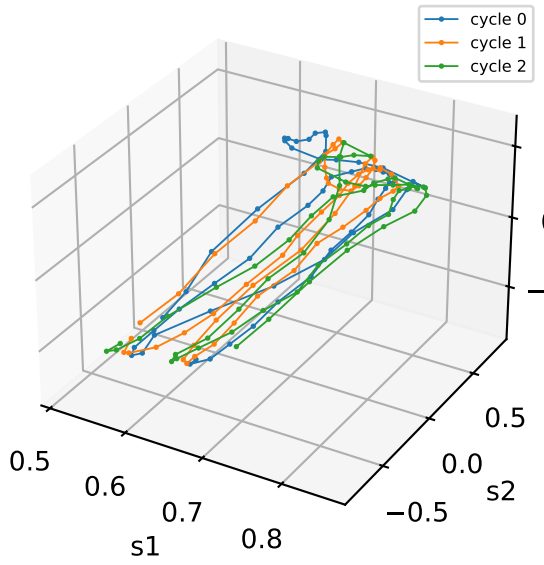


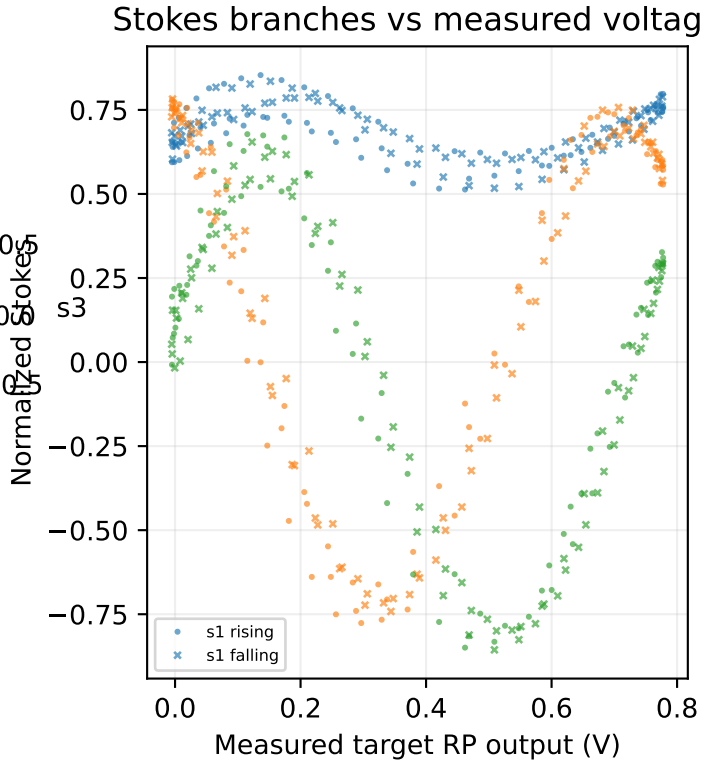


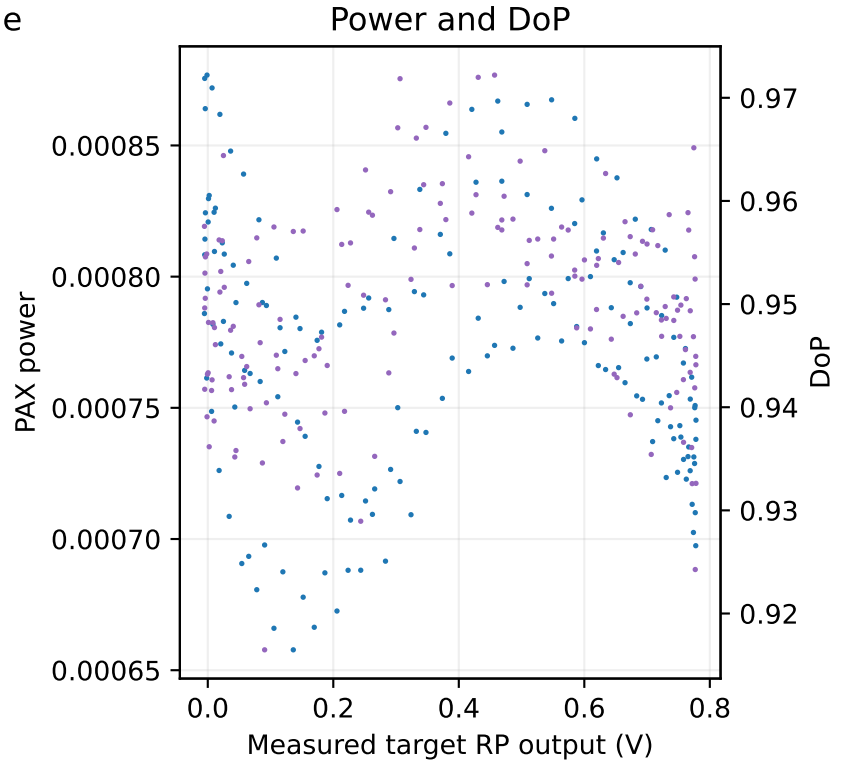


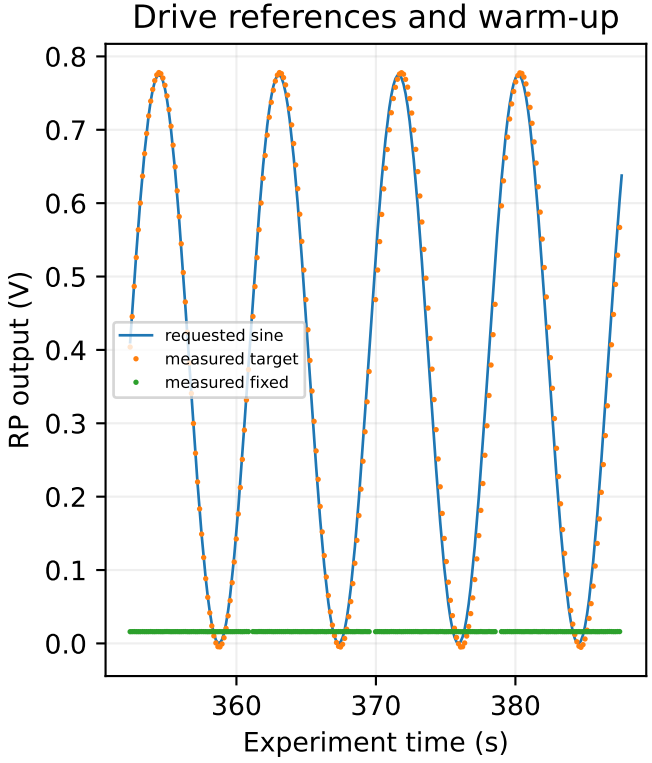
Ideal reference: phi1 should keep s1 fixed and total power independent of drive.
Plane normal: [0.989, -0.038, -0.140]
Plane residual RMS: 0.02902
Rising/falling angular RMS: 8.29°
Rising/falling power RMS: 4.35e-05 PAX units
Cycle angular RMS: 8.78°
Cycle power RMS: 5.72e-05 PAX units
Cycle DoP RMS: 0.0182
Tracking RMS: 0.03309 V
Reference coverage: 97.2%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.98, -0.00, -0.21], residual 0.00579
Cycle 1 plane: [0.98, -0.01, -0.20], residual 0.0194
Cycle 2 plane: [0.99, -0.07, -0.10], residual 0.0274

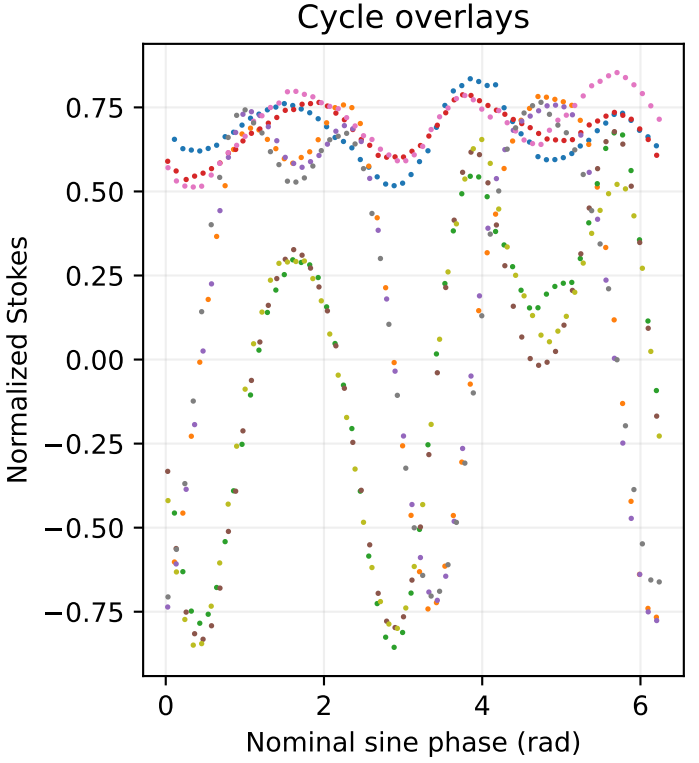
phi1 sine; fixed phi2 = 0.62 RP V (segment 8)





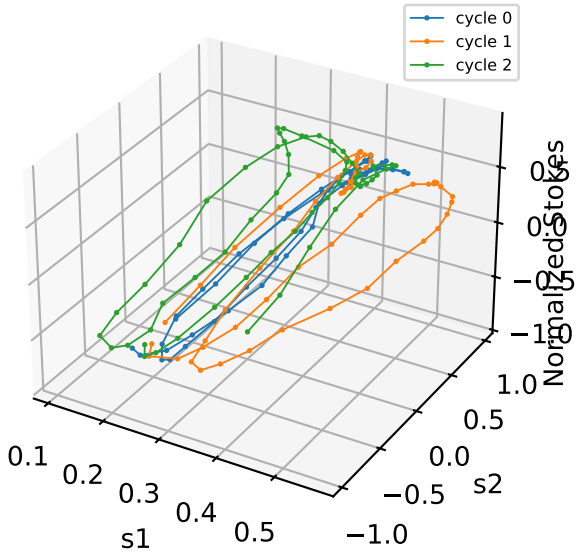


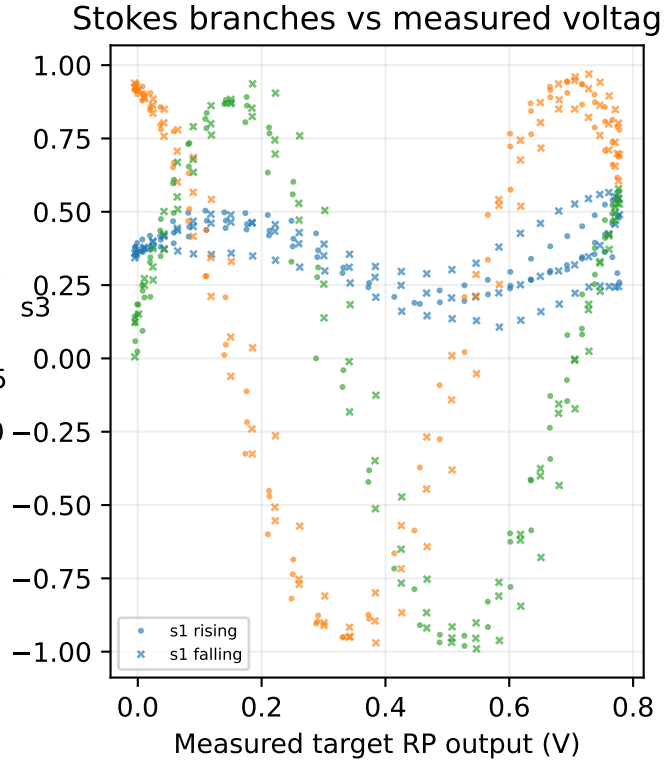


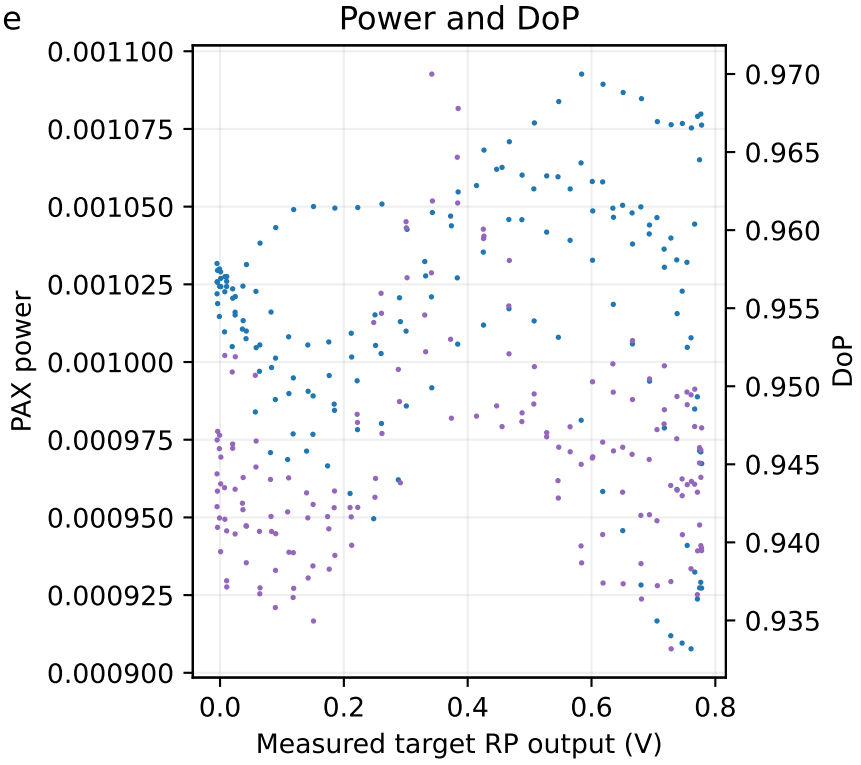


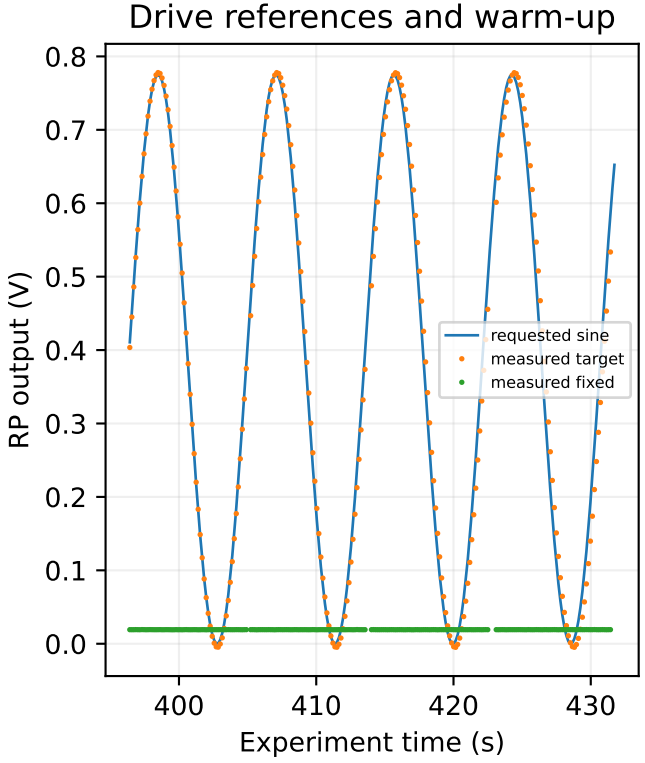
Ideal reference: phi1 should keep s1 fixed and total power independent of drive.
Plane normal: [0.990, 0.004, -0.143]
Plane residual RMS: 0.04639
Rising/falling angular RMS: 12.5°
Rising/falling power RMS: 6.6e-05 PAX units
Cycle angular RMS: 6.68°
Cycle power RMS: 6.39e-05 PAX units
Cycle DoP RMS: 0.00804
Tracking RMS: 0.03187 V
Reference coverage: 97.2%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.99, 0.02, -0.13], residual 0.0536
Cycle 1 plane: [0.99, -0.01, -0.12], residual 0.03
Cycle 2 plane: [0.98, 0.01, -0.18], residual 0.0421

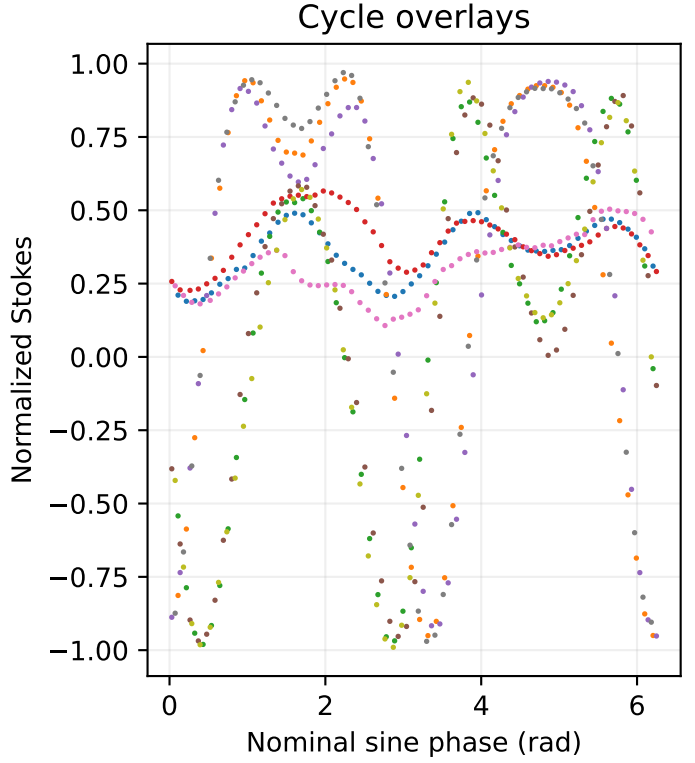
phi1 sine; fixed phi2 = 0.6974 RP V (segment 9)





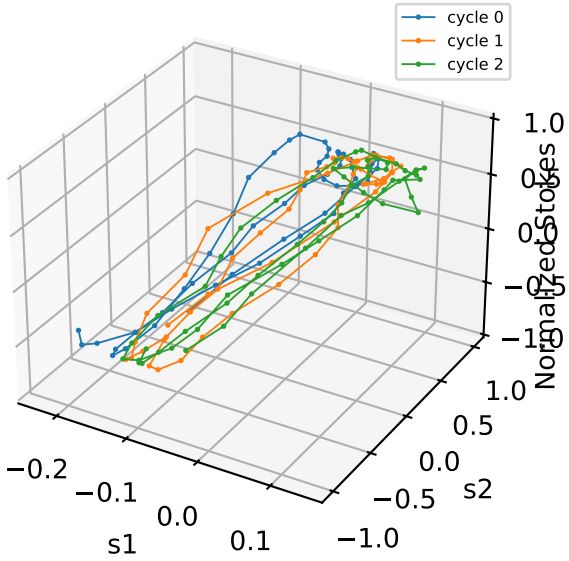




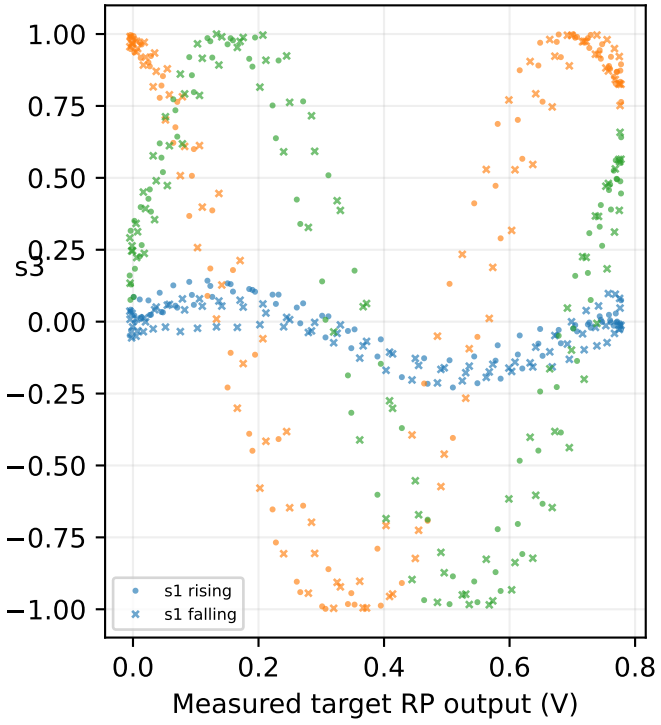


Ideal reference: phi1 should keep s1 fixed and total power independent of drive.
Plane normal: [0.991, -0.019, -0.136]
Plane residual RMS: 0.06702
Rising/falling angular RMS: 15.2°
Rising/falling power RMS: 3.88e-05 PAX units
Cycle angular RMS: 9.41°
Cycle power RMS: 4.52e-05 PAX units
Cycle DoP RMS: 0.00362
Tracking RMS: 0.03 V
Reference coverage: 96.6%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.99, -0.02, -0.15], residual 0.0204
Cycle 1 plane: [0.99, -0.05, -0.11], residual 0.0587
Cycle 2 plane: [0.99, -0.00, -0.14], residual 0.0666

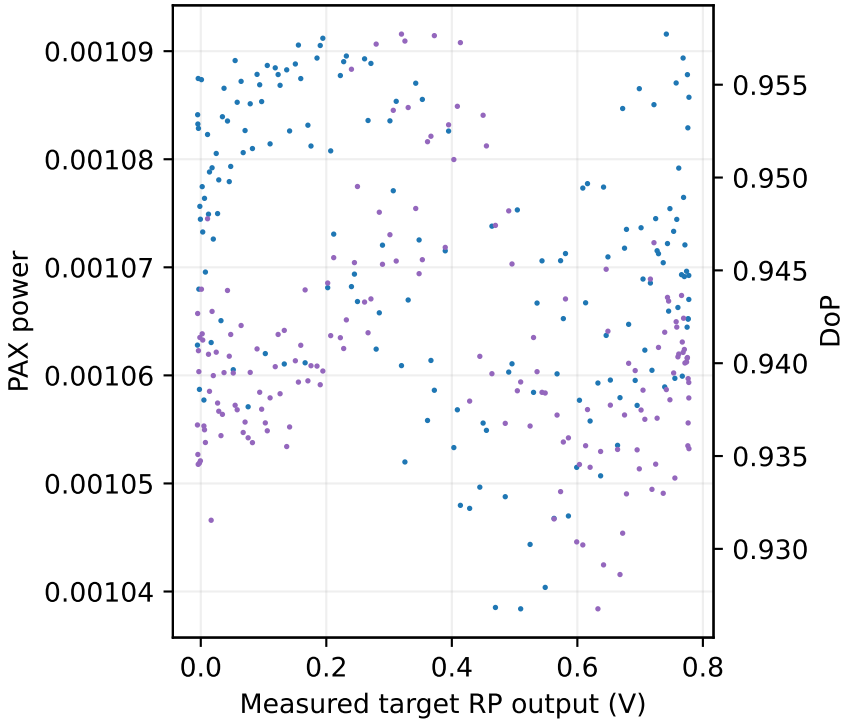
phi1 sine; fixed phi2 = 0.7749 RP V (segment 10)



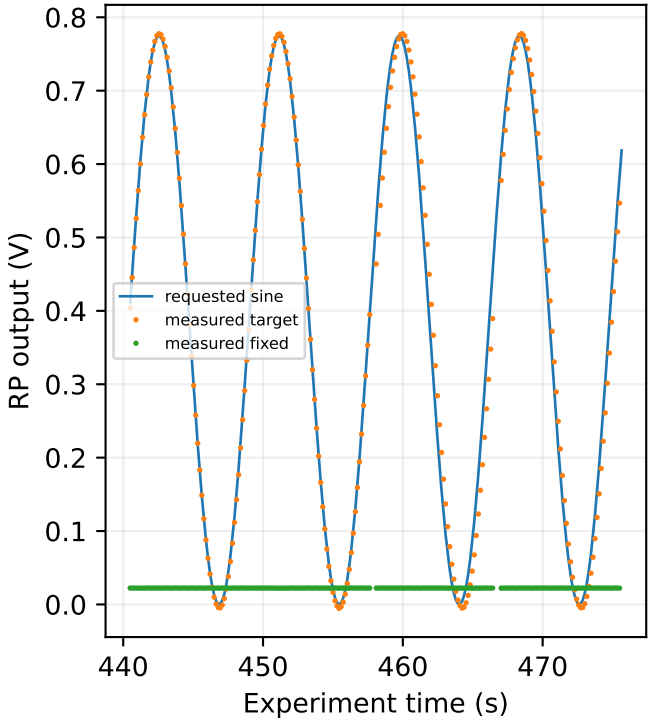
Stokes branches vs measured voltage



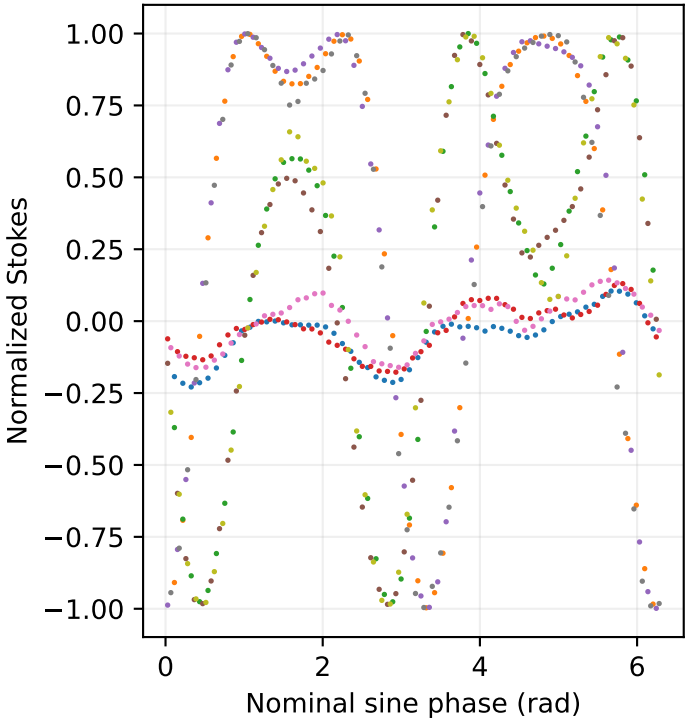
Power and DoP



Drive references and warm-up



Cycle overlays

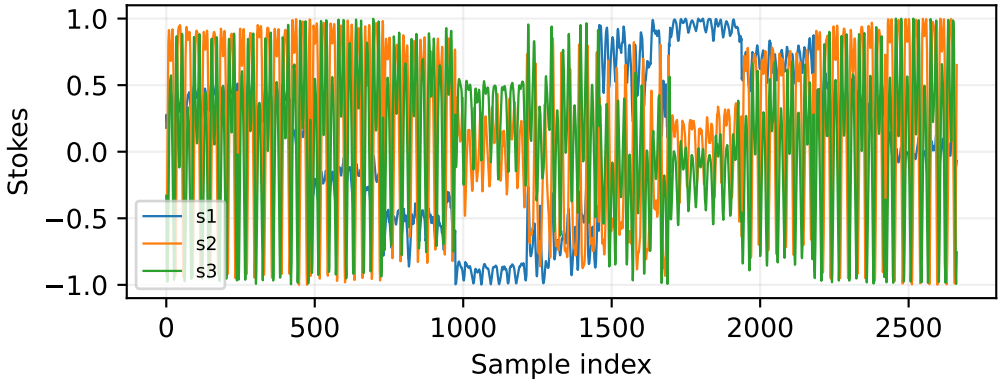


Ideal reference: phi1 should keep s1 fixed and total power independent of drive.
Plane normal: [0.992, -0.015, -0.127]
Plane residual RMS: 0.0389
Rising/falling angular RMS: 24°
Rising/falling power RMS: 1.49e-05 PAX units
Cycle angular RMS: 6.6°
Cycle power RMS: 1.57e-05 PAX units
Cycle DoP RMS: 0.00348
Tracking RMS: 0.0311 V
Reference coverage: 97.2%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.99, -0.01, -0.13], residual 0.034
Cycle 1 plane: [0.99, -0.01, -0.12], residual 0.0316
Cycle 2 plane: [0.99, -0.02, -0.13], residual 0.032

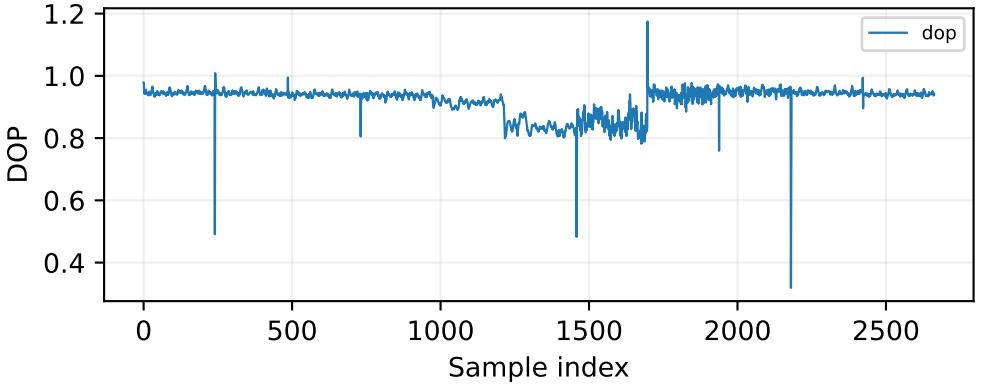
Recorded telemetry

Acquisition order preserves stage changes; no timing alignment or transfer gain is inferred.

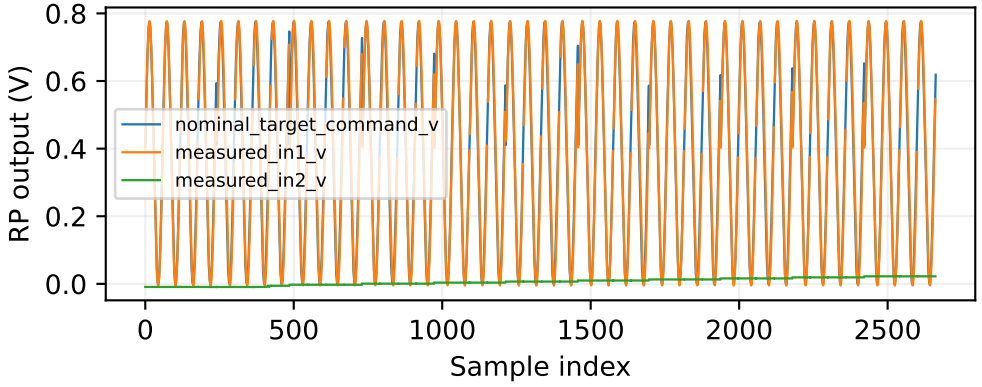
Normalized Stokes



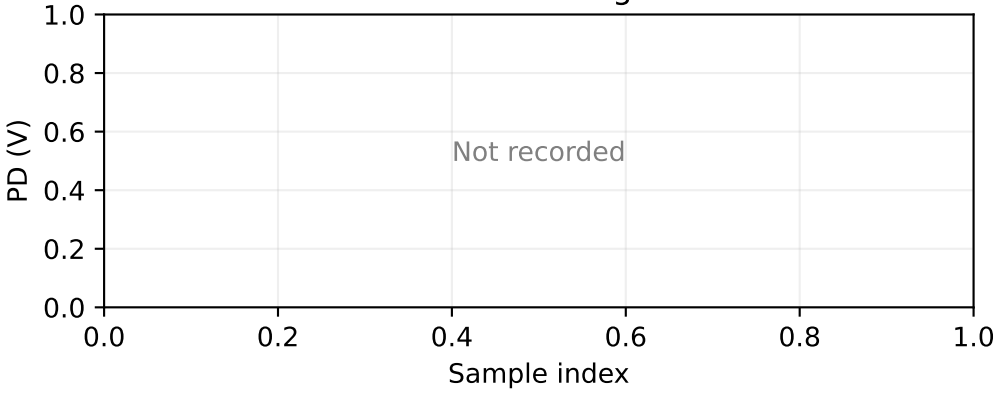
Raw degree of polarization



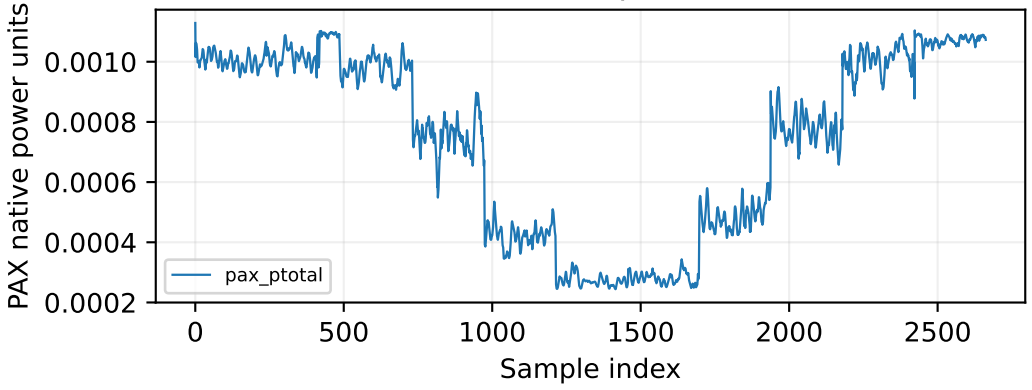
Actuator command / reference history



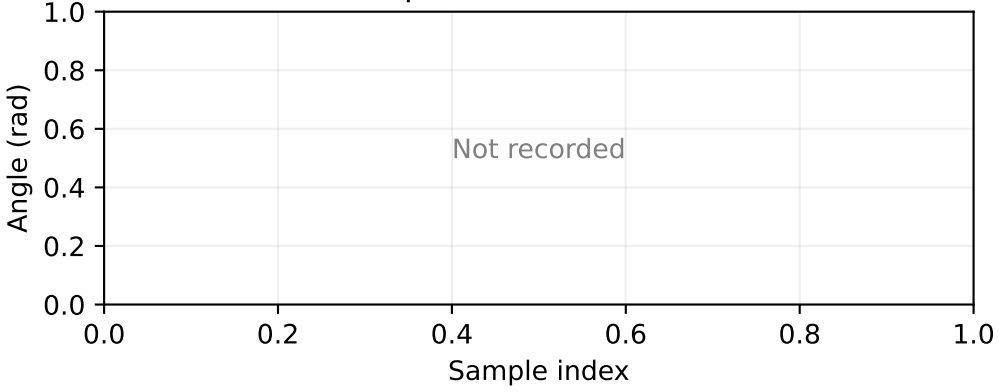
Photodiode signal



PAX total power



Sphere coordinates



Active parameters

RP commands: volts, bounded to 0–1. V_lambda: terminal volts for 2π. Gains: terminal V / RP V.

Parameter	Recorded value
label	"quick-test"
comment	"cold run, haven't tested this tool update yet. this is less of a po
actuator	"phi1"
report	"pdf"
pax_rotation_velocity_hz	60.0
pax_measurement_mode	5
pax_wavelength_nm	830.0
phi1_v_lambda	13.078125
phi2_v_lambda	13.078125
phi1_actuator_volts_per_rp_volt	16.875
phi2_actuator_volts_per_rp_volt	16.875
rp_output_min_voltage	0.0
rp_output_max_voltage	1.0
cross_sweep_sine_frequency_hz	0.1
cross_sweep_sine_center_voltage	0.3875
cross_sweep_sine_amplitude_voltage	0.3875
cross_sweep_warmup_cycles	1
cross_sweep_recorded_cycles	3
cross_sweep_phi1_bias_voltages	[0.0, 0.0775, 0.155, 0.2325, 0.31, 0.3875, 0.465, 0.5425, 0.62, 0
cross_sweep_phi2_bias_voltages	[0.0, 0.0775, 0.155, 0.2325, 0.31, 0.3875, 0.465, 0.5425, 0.62, 0
cross_sweep_settle_s	1.0
cross_sweep_bias_ramp_s	1.0
cross_sweep_bias_ramp_updates_per_s	50.0
cross_sweep_reference_sample_rate_hz	400.0

Active parameters

RP commands: volts, bounded to 0–1. V_lambda: terminal volts for 2π. Gains: terminal V / RP V.

Parameter	Recorded value
cross_sweep_scope_block_s	8.0
cross_sweep_min_pax_samples_per_cycle	20.0